

Lazarus - Management Plan

astroFUEL

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Acronyms

AIT Assembly, Integration and Testing.

CDR Critical Design Review.

ECSS European Cooperation for Space Standardization.

EOL End of Life.

EOM End of Mission.

EP Electric Propulsion.

ESA European Space Agency.

FAR Flight Acceptance Review.

FEM Finite Element Method.

FSW Flight Software.

GEO Geostationary Orbit.

ICD Interface Control Documents.

LEOP Launch and Early Orbit Phase.

MDR Mission Definition Review.

MLI Multi-layer insulation.

PBS Project Breakdown Structure.

PDR Preliminary Design Review.

PLA Payload Adapter.

PRR Preliminary Requirements Review.

QR Qualification Request.

SRR System Requirements Review.

WBS Work Breakdown Structure.

WP Work Package.

1 Introduction

Purpose of the document, mission overview and scope.

LAZARUS is focused on demonstrating an in-orbit refuelling service for satellites in geostationary orbit. The mission requires the coordination of several technical areas working together on the spacecraft, the refuelling payload, mission operations and launcher-related activities.

For this reason, a clear management structure is required to define responsibilities, organize the work, establish the main project phases and plan the activities needed throughout the development. This Management Plan provides the common framework used by the different technical areas to coordinate their work and follow the evolution of the project.

The document presents the internal organization of astroFUEL for the LAZARUS mission, the Project Breakdown Structure (PBS), the project phasing, the project planning, the main project risks and the Work Breakdown Structure (WBS). Both the preliminary design stage and the complete project lifecycle are considered, providing a general view of how the mission is expected to progress from its initial definition to operations and End of Life.

2 Project Organization

2.1 Identification of the customer

Market Definition: 2030s-2040s GEO Landscape

Propellant exhaustion is the principal limitation in the design life of a satellite and the primary barrier to unlocking immense financial value in geostationary orbit. An estimated one satellite out of roughly five in Geostationary Orbit (GEO) is shut down just because it ran out of station keeping fuel, even though its electronics, transponders, and solar panels still work perfectly. LAZARUS overcomes this limitation, in particular it sits at the intersection of modern satellite design and strict space sustainability rules. Our main market includes high capacity commercial telecom satellites and secure government defense assets. These operate in Geostationary Orbit using Xenon based Electric Propulsion (EP). The roughly 2030s and estimated 2040s are the perfect time for our business. Driven by strict zero debris regulations from The European Space Agency (ESA) and others, and the current improvement on in orbit servicing technologies, major manufacturers like Airbus and Thales Alenia Space are making a significant shift. They will now include standard servicing ports on their satellites before launch. We are not building a complex, high risk robotic tow truck to chase down old debris. Instead, LAZARUS is a simple, plug and play refueling service built for this new generation of cooperative satellites. It provides a low risk, cost effective alternative to replacing expensive assets. This creates a clear and actionable market:

- Total Addressable Market: The entire global fleet of commercial and defense satellites operating in GEO during the roughly 2030s and estimated 2040s.
- Serviceable Available Market: The growing segment of next generation GEO satellites equipped with Xenon EP systems and standardized servicing ports.
- Serviceable Obtainable Market: The initial roughly 5 to an estimated 10 high capacity satellites managed by European commercial and defense operators that we will target for our first campaign.

Customer Profiles and Fleet Alignment

AstroFUEL targets operators who manage valuable space assets weighing between roughly 1,000 kg and an estimated 6,500 kg at launch. Our ideal customers are proactive fleet planners, not operators dealing with an emergency. They manage multibillion dollar fleets where keeping satellites alive longer directly boosts their return on investment. We divide this customer base into two main profiles:

- Commercial Fleet Operators: Tier 1 companies like Eutelsat Group, SES, Hispasat, Intelsat, and Viasat. Their main goal is to keep their fully paid off satellites generating lease revenue for longer. Recent industry orders, like SES contracting Thales Alenia Space for the Space Inspire and Spacebus Neo innovative product lines, and Intelsat buying from SWISSto12, show heavy investment in Xenon reliant designs.

- Government and Defense Organizations: Institutions like Hisdesat, the French Directorate General of Armaments, and allied defense ministries. For these clients, keeping their secure payloads online and maintaining orbital control is more important than standard corporate retirement schedules.

Customer qualification comes down to hardware compatibility. LAZARUS is designed to connect directly with the main satellite buses adopting standard refueling ports:

- Heavy and High Capacity Platforms, roughly 3,400 to more than an estimated 6,500 kg: Systems like the Airbus Eurostar Neo and Thales Alenia Space Spacebus Neo. This includes flagship programs like Hisdesat's Spainsat NG I and II, the UK's Skynet 6A, and major commercial payloads.
- Software Defined Architectures, approximately 4,500 kg: Mid to heavy platforms like the Thales Alenia Space Space Inspire line. These support reprogrammable missions that benefit greatly from extra time in orbit.
- SmallGEO Platforms, roughly 1,000 kg: Newer, smaller satellite designs that use electric propulsion, such as SWISSto12's HummingSat.

Operational Sizing and Contracting Strategy

LAZARUS carries a payload of approximately 500 kg of Xenon gas. The economic rationale for our service is driven by the massive potential annual gain unlocked for every year of extended life we provide. According to an MIT study, transponder lease rates average an estimated 5,155 dollars per megahertz per month for Ku band capacity and roughly 4,921 dollars per megahertz per month for C band capacity. By extending active service, operators can generate a potential commercial gain approaching 50 million dollars for every roughly 6 months of operation, or upwards of 100 million dollars for every estimated single year of additional life. Since fuel consumption scales with the size of the satellite, we map our payload directly to specific bus architectures. Heavy assets like the Airbus Eurostar Neo and Thales Alenia Space Spacebus Neo burn roughly 15 to an estimated 21 kg of Xenon a year. Mid class software defined platforms like the Space Inspire line use approximately 11 to around 15 kg, and SmallGEOs like the HummingSat need just roughly 4 to an estimated 5 kg. Based on these consumption rates, we offer two distinct services tailored to European anchor operators:

- Deep Life Extension: We can deliver roughly 100 kg of Xenon to an estimated 5 heavy satellites. Targeting Tier 1 commercial operators like Eutelsat Group or SES managing massive Spacebus Neo or Eurostar Neo platforms, this service adds approximately 5 to roughly 7 years of revenue generating life to each asset. This generates upwards of hundreds of millions of dollars in new potential gain per satellite and lets operators delay the massive cost of launching replacements.
- Broad Fleet Coverage: We can distribute roughly 50 kg across an estimated 10 mid class or defense satellites. Targeting sovereign clients like Hisdesat, this service adds an estimated

3 to roughly 4 years of operational service to each platform, giving fleet managers more flexibility and mission continuity.

AstroFUEL works as a long term planning partner. Customers sign Service Level Agreements approximately two to roughly three years before they run out of fuel. This gives us predictable, recurring revenue and lets operators easily build refueling into their long term corporate forecasts.

The Early Adopter Strategy

In the roughly 2030s, replacing a perfectly good satellite too early simply will not make financial sense. LAZARUS provides a direct solution. With a low risk, fully autonomous system backed by a strong European foundation, it is a service that practically pays for itself. We are focusing first on European leaders like Eutelsat Group, SES, and Hisdesat to execute the operational scenarios outlined above. Getting these early adopters on board will prove our system works and make planned refueling the industry standard. Our first clients will be forward thinking operators who implement standard refueling ports in their satellites, since they know that planning today guarantees their fleet's profitability tomorrow.

2.2 Company Organization

AstroFUEL is a private initiative in the European new space market. The company manages the complete spacecraft development process in-house, from mission analysis and subsystem design through manufacturing, integration, and final assembly. This approach provides full control over the spacecraft development process and allows the company to make decisions independently, while also enabling it to respond rapidly to new business opportunities arising in the constantly evolving new space market.

External suppliers are used exclusively for the procurement of the materials, components, and parts required for the spacecraft. The integration and assembly of these elements into the spacecraft are carried out entirely in-house by astroFUEL.

To enable fast and effective decision-making, the company follows a clear chain of command. The CEO is the head of the company and oversees all ongoing projects, allocating the available company resources to each project according to its needs. The CEO appoints a Project Manager for each project, who is responsible for initiating, planning, executing, monitoriong, controlling and closing the project. This includes scope, time, cost, quality, risk and human resource management.

Under the responsibility of the Project Manager, the technical development of each project is divided into different areas. Since astroFUEL's first project, LAZARUS, is for now only a technology demonstrator at a preliminary design stage, the company has only technical areas, pending the move to operate as a commercial enterprise. That is when the legal, HR, commercial, and environmental departments, among others, will be established.

The technical development of LAZARUS is divided into the following areas: Propulsion, Mission Analysis, Mechanical & Structural Engineering, Communications & Data Handling, Thermal Management, Payload, Attitude & Orbit Control/Guidance, Navigation & Control, and Power Management (see Figure 2.1). Each technical area is responsible for the analysis, design, and development of its corresponding subsystem or technical domain, as well as for contributing to the integration of the complete spacecraft.

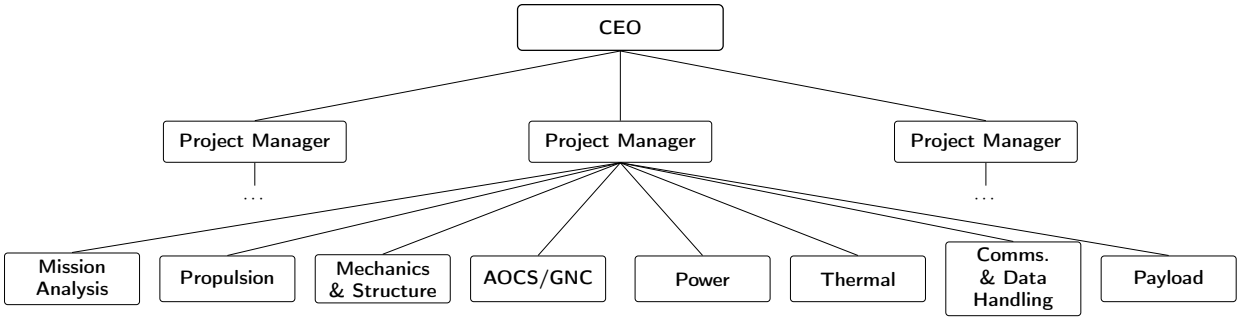


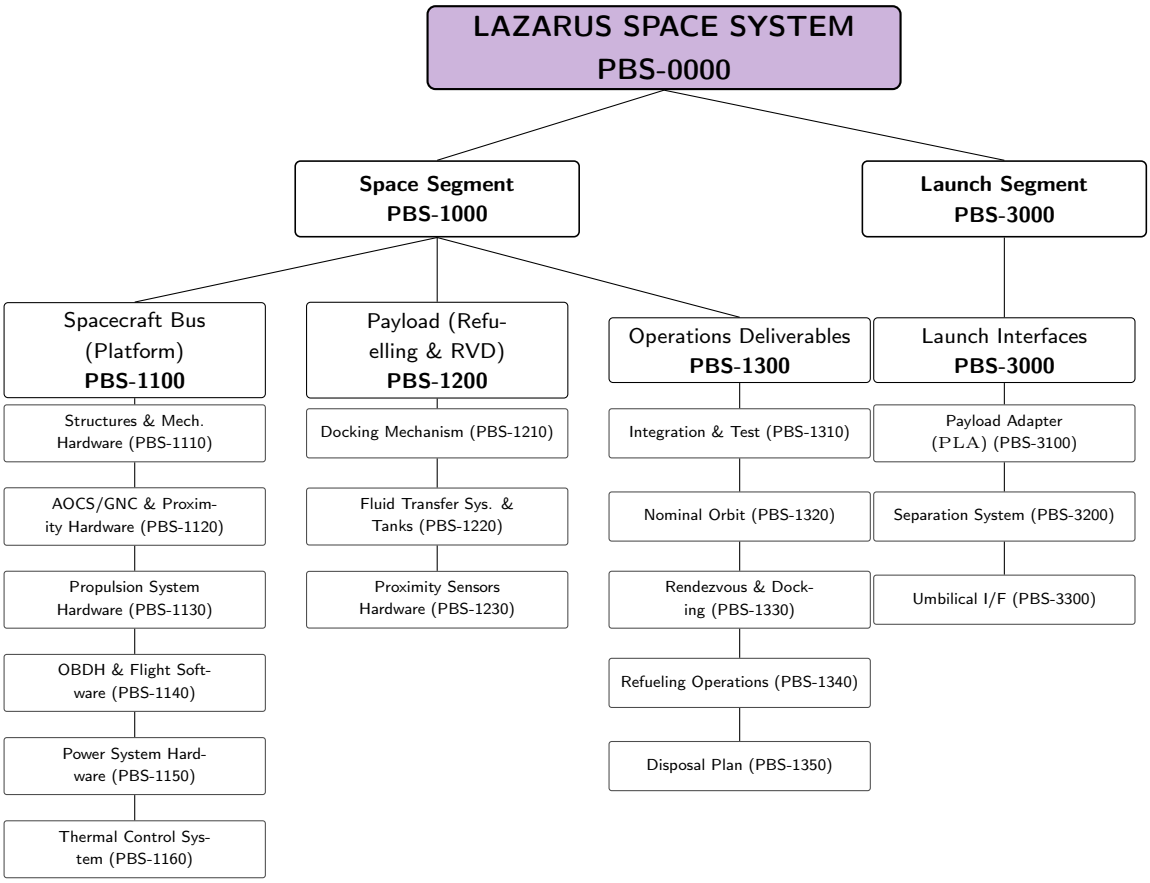
Figure 2.1: Schematic representation of the astroFUEL's internal company organization.

Mission Analysis is responsible for defining and analysing the mission scenarios and orbital requirements. Propulsion is responsible for the design and development of the propulsion system required for spacecraft manoeuvring. Attitude & Orbital Control/Guidance, Navigation & Control (AOCS/GNC) is responsible for the attitude and orbit control functions required throughout the mission, including maintaining the target orbits and facing and performing the operations required to approach and service the client satellite. Mechanical & Structural Engineering is responsible for the mechanical design of the spacecraft, including the primary and secondary structures, as well as for the integration of all subsystems into the spacecraft structure. Power Management is responsible for the generation, storage, and distribution of electrical power throughout the spacecraft, ensuring that all subsystems can operate nominally throughout the mission. Thermal Management is responsible for the thermal design and analysis of the spacecraft, ensuring that all components and subsystems remain within their specified operating temperature ranges. Particular attention is given to the payload, which requires specific thermal conditions throughout the mission. Communications & Data Handling is responsible for the spacecraft's communication functions, onboard data handling, and data transmission. Finally, the Payload area is responsible for coordinating with the other technical areas the design and development of the equipment required to store and handle Xenon under the specified conditions throughout the mission, as well as to perform the servicing and refuelling of client satellites.

3 PBS

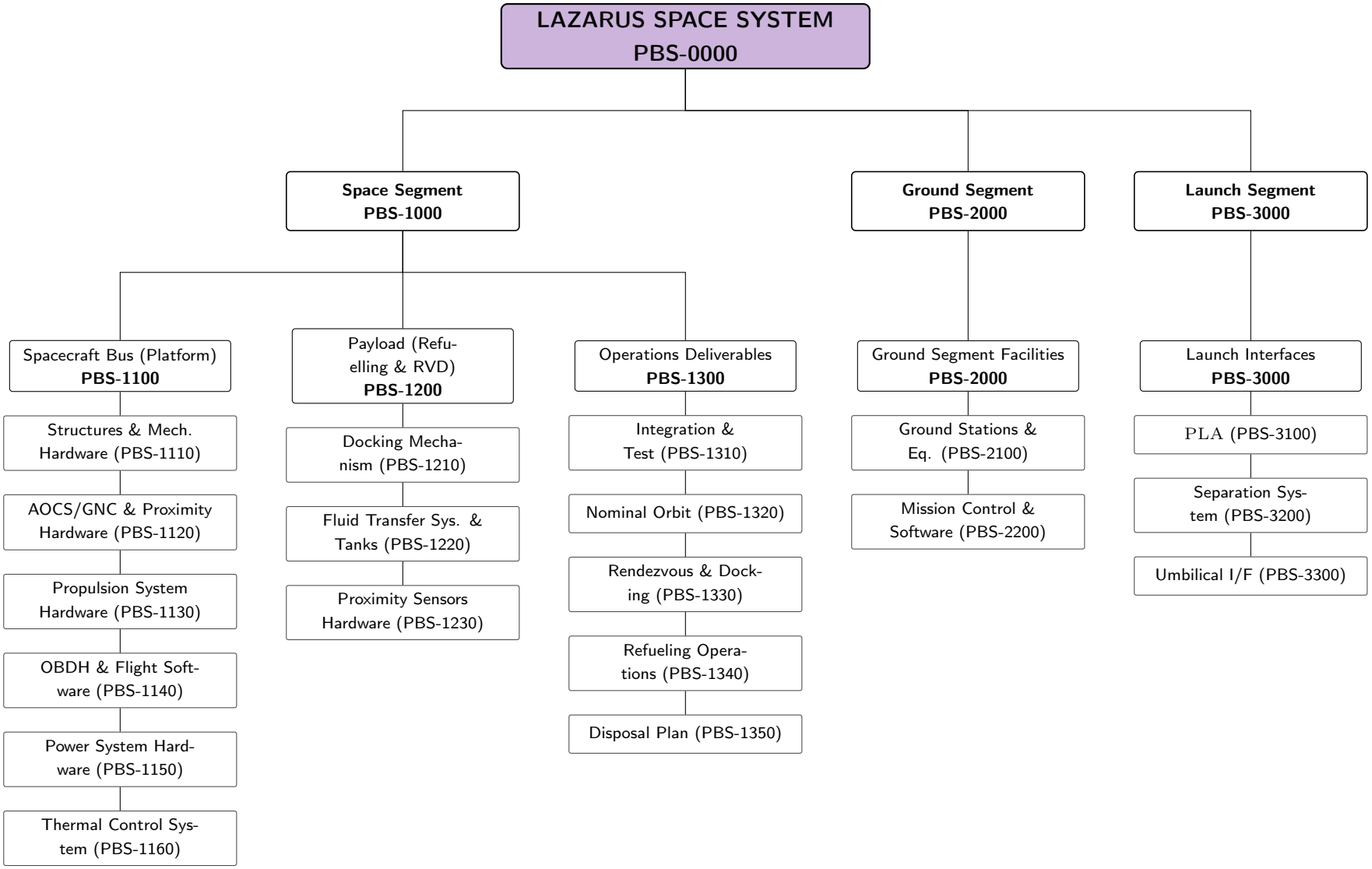
3.1 PBS of the Preliminary Design Project

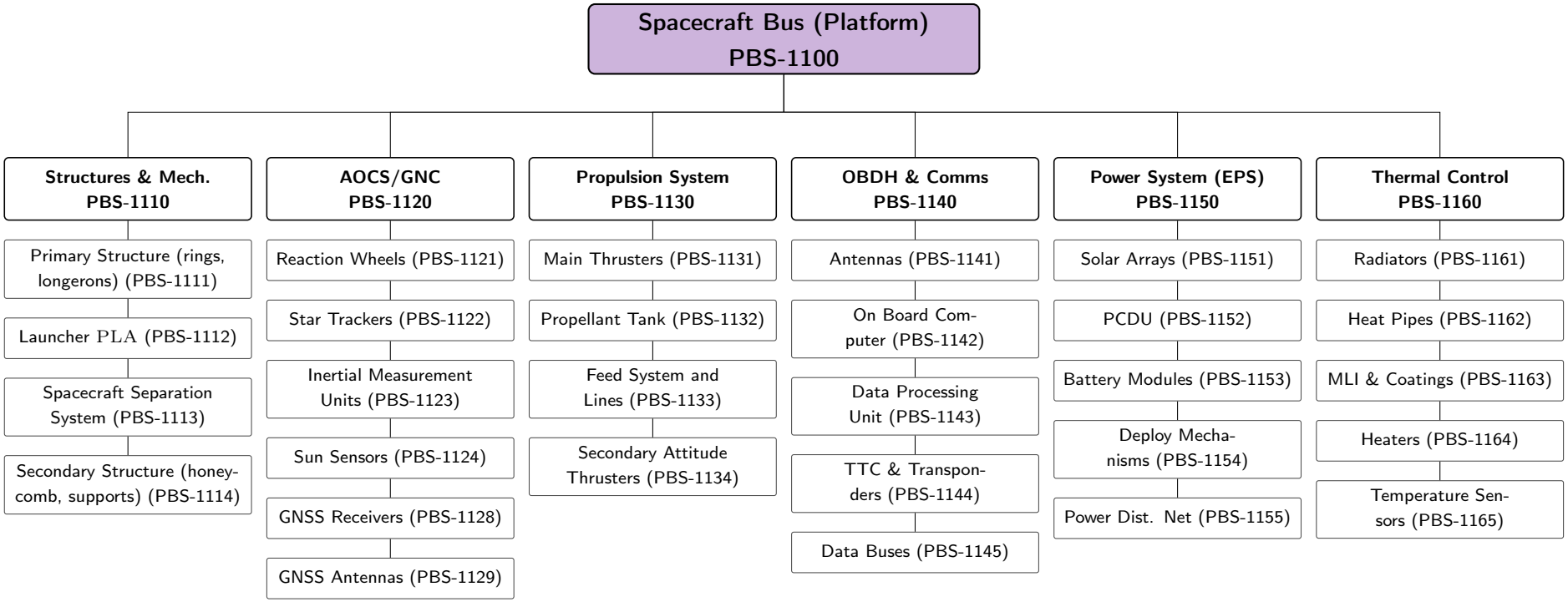
The Preliminary Design PBS outlines the physical and document deliverables required during the initial study and design phases (Phases 0, A, and B). At this stage, the structure focuses strictly on defining the core mission architecture: the Spacecraft Bus, the Refueling Payload, preliminary Operations Deliverables, and early Launch Interfaces. The Ground Segment is deliberately excluded from this preliminary tree, as its detailed development, validation, and procurement are scheduled to begin later in Phase C, ensuring strict alignment with the project's WBS and phasing.

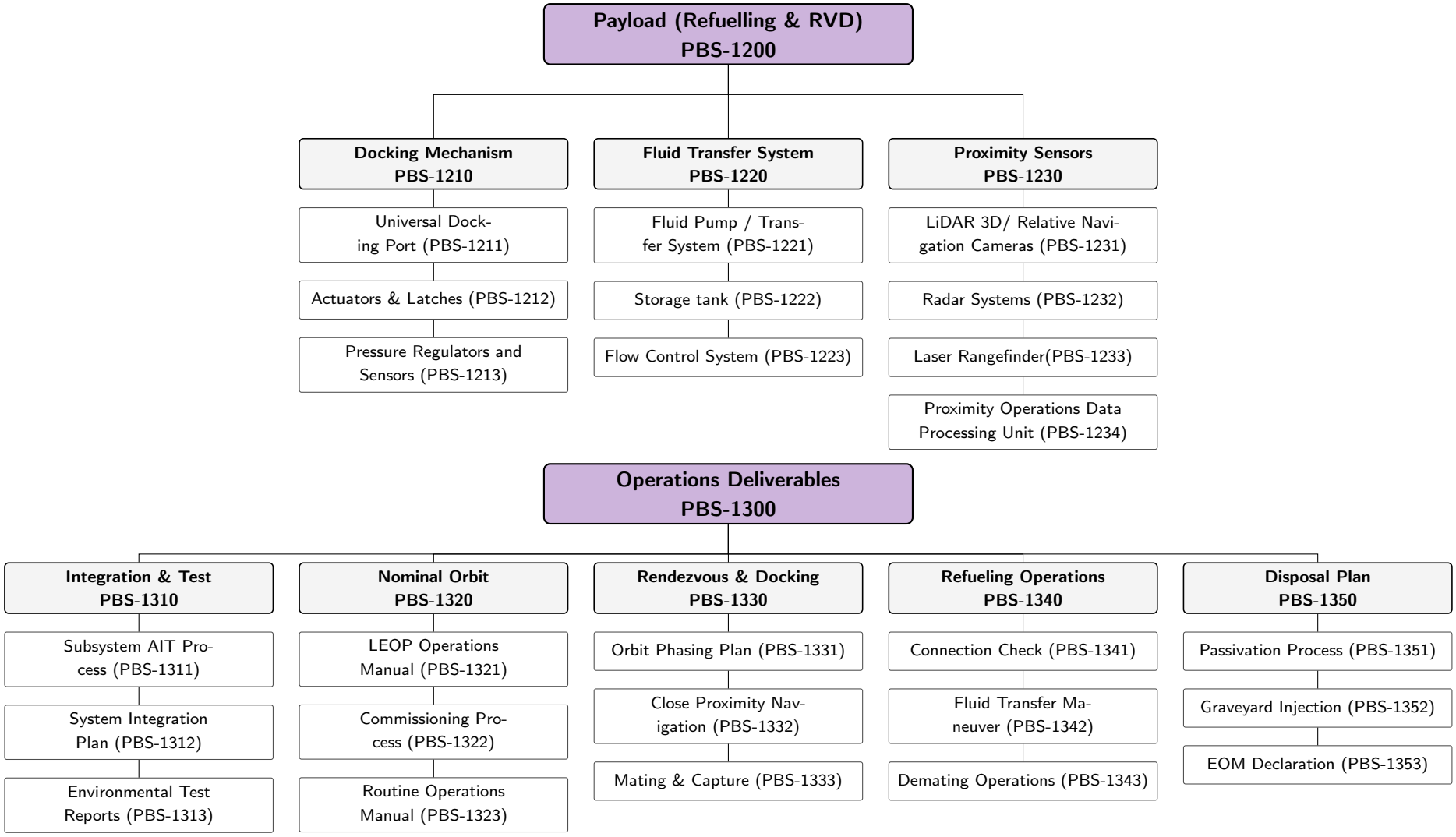


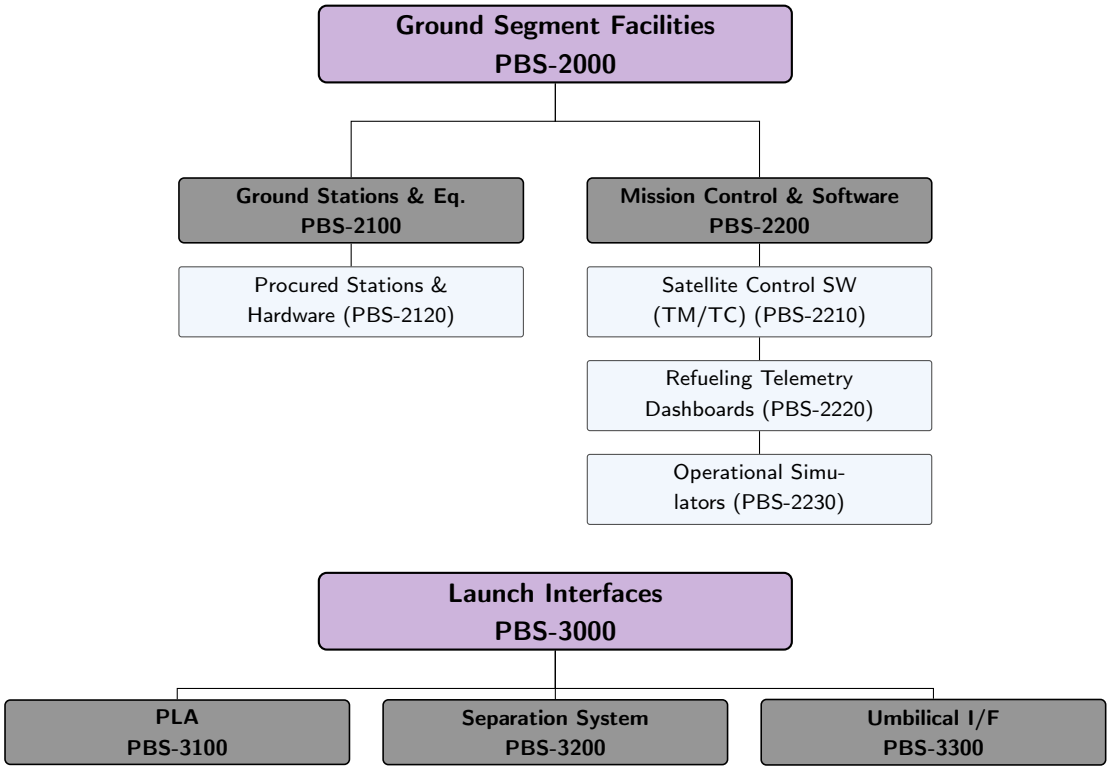
3.2 PBS of the complete Project

The Complete Project expands upon the preliminary foundation to represent the entire LAZARUS mission lifecycle, extending through End of Life (EOL) disposal (Phase F). It integrates the Ground Segment Facilities, including Ground Stations and Mission Control Software, and provides a comprehensive, lower-level breakdown of all major segments.


LAZARUS SPACE SYSTEM
PBS-0000
Space Segment
PBS-1000
Ground Segment
PBS-2000
Launch Segment
PBS-3000
Spacecraft Bus (Platform)
PBS-1100
Structures & Mech. Hardware (PBS-1110)
AOCS/GNC & Proximity Hardware (PBS-1120)
Propulsion System Hardware (PBS-1130)
OBDH & Flight Software (PBS-1140)
Power System Hardware (PBS-1150)
Thermal Control System (PBS-1160)
Payload (Refuelling & RVD)
PBS-1200
Docking Mechanism (PBS-1210)
Fluid Transfer Sys. & Tanks (PBS-1220)
Proximity Sensors Hardware (PBS-1230)
Operations Deliverables
PBS-1300
Integration & Test (PBS-1310)
Nominal Orbit (PBS-1320)
Rendezvous & Docking (PBS-1330)
Refueling Operations (PBS-1340)
Disposal Plan (PBS-1350)
Ground Segment Facilities
PBS-2000
Ground Stations & Eq. (PBS-2100)
Mission Control & Software (PBS-2200)
Launch Interfaces
PBS-3000
PLA (PBS-3100)
Separation System (PBS-3200)
Umbilical I/F (PBS-3300)


Spacecraft Bus (Platform)
PBS-1100
Structures & Mech.
PBS-1110
Primary Structure (rings, longerons) (PBS-1111)
Launcher PLA (PBS-1112)
Spacecraft Separation System (PBS-1113)
Secondary Structure (honeycomb, supports) (PBS-1114)
AOCS/GNC
PBS-1120
Reaction Wheels (PBS-1121)
Star Trackers (PBS-1122)
Inertial Measurement Units (PBS-1123)
Sun Sensors (PBS-1124)
GNSS Receivers (PBS-1128)
GNSS Antennas (PBS-1129)
Propulsion System
PBS-1130
Main Thrusters (PBS-1131)
Propellant Tank (PBS-1132)
Feed System and Lines (PBS-1133)
Secondary Attitude Thrusters (PBS-1134)
OBDH & Comms
PBS-1140
Antennas (PBS-1141)
On Board Computer (PBS-1142)
Data Processing Unit (PBS-1143)
TTC & Transponders (PBS-1144)
Data Buses (PBS-1145)
Power System (EPS)
PBS-1150
Solar Arrays (PBS-1151)
PCDU (PBS-1152)
Battery Modules (PBS-1153)
Deploy Mechanisms (PBS-1154)
Power Dist. Net (PBS-1155)
Thermal Control
PBS-1160
Radiators (PBS-1161)
Heat Pipes (PBS-1162)
MLI & Coatings (PBS-1163)
Heaters (PBS-1164)
Temperature Sensors (PBS-1165)


Payload (Refuelling & RVD)
PBS-1200
Docking Mechanism
PBS-1210
Universal Docking Port (PBS-1211)
Actuators & Latches (PBS-1212)
Pressure Regulators and Sensors (PBS-1213)
Fluid Transfer System
PBS-1220
Fluid Pump / Transfer System (PBS-1221)
Storage tank (PBS-1222)
Flow Control System (PBS-1223)
Proximity Sensors
PBS-1230
LiDAR 3D/ Relative Navigation Cameras (PBS-1231)
Radar Systems (PBS-1232)
Laser Rangefinder(PBS-1233)
Proximity Operations Data Processing Unit (PBS-1234)
Operations Deliverables
PBS-1300
Integration & Test
PBS-1310
Subsystem AIT Process (PBS-1311)
System Integration Plan (PBS-1312)
Environmental Test Reports (PBS-1313)
Nominal Orbit
PBS-1320
LEOP Operations Manual (PBS-1321)
Commissioning Process (PBS-1322)
Routine Operations Manual (PBS-1323)
Rendezvous & Docking
PBS-1330
Orbit Phasing Plan (PBS-1331)
Close Proximity Navigation (PBS-1332)
Mating & Capture (PBS-1333)
Refueling Operations
PBS-1340
Connection Check (PBS-1341)
Fluid Transfer Maneuver (PBS-1342)
Demating Operations (PBS-1343)
Disposal Plan
PBS-1350
Passivation Process (PBS-1351)
Graveyard Injection (PBS-1352)
EOM Declaration (PBS-1353)


Ground Segment Facilities
PBS-2000
Ground Stations & Eq.
PBS-2100
Mission Control & Software
PBS-2200
Procured Stations &
Hardware (PBS-2120)
Satellite Control SW
(TM/TC) (PBS-2210)
Refueling Telemetry
Dashboards (PBS-2220)
Operational Simu-
lators (PBS-2230)
Launch Interfaces
PBS-3000
PLA
PBS-3100
Separation System
PBS-3200
Umbilical I/F
PBS-3300

4 Project Phasing

LAZARUS is divided into a series of sequenced and consecutive phases covering the entire mission's lifecycle, from the initial definition of the mission to the end of the spacecraft's operational lifetime. As outlined in Figure 4.1, each phase is composed of a number of tasks (white boxes) and milestones (purple boxes).

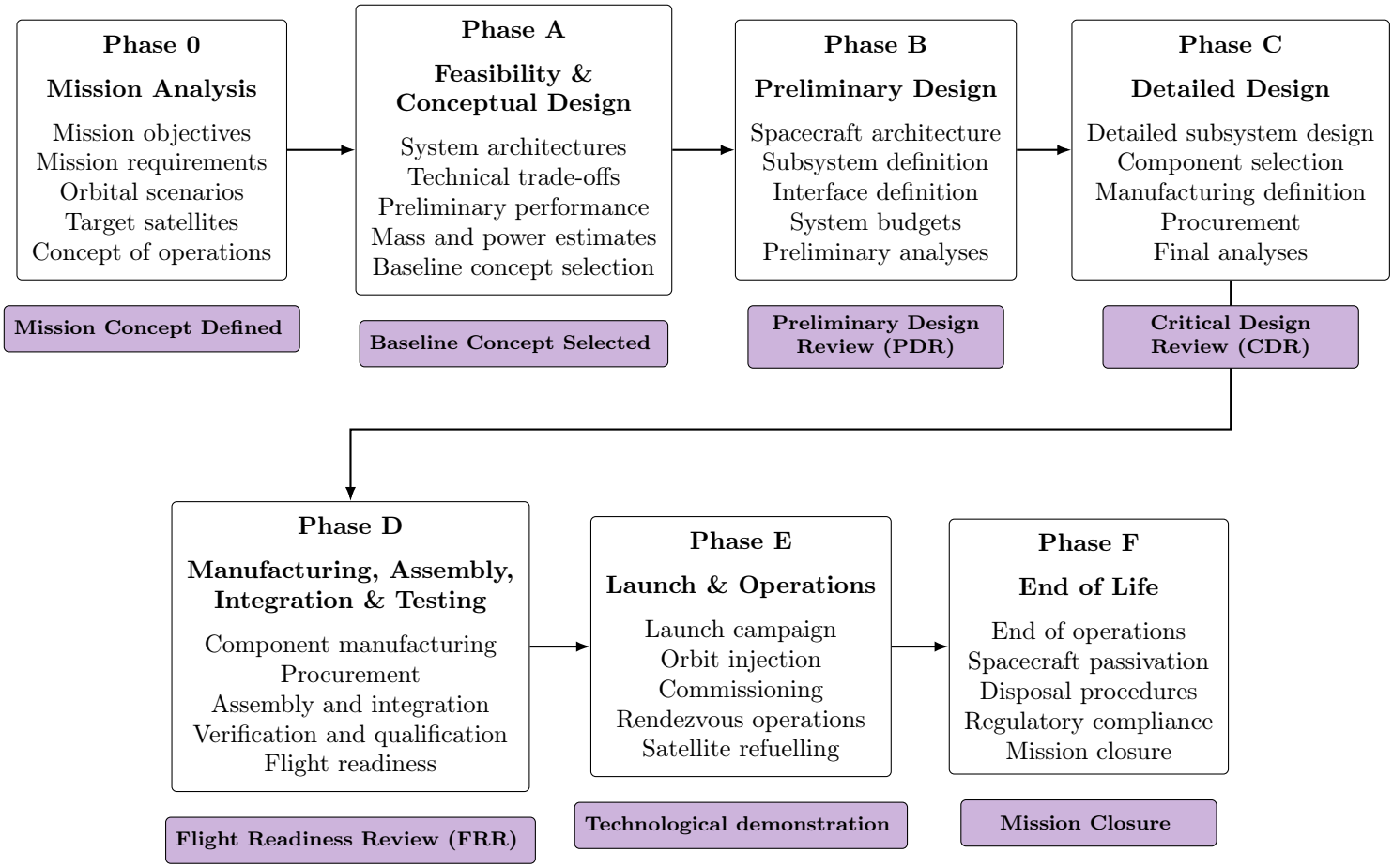


Figure 4.1: LAZARUS project phases, main activities and milestones.

The main project phases are defined as follows:

Phase 0 – Mission Analysis

During this phase, the mission objectives and concept of operations are defined. The main mission requirements, orbital scenarios, target satellites, and overall concept for the in-orbit refuelling mission are established. Initial feasibility assessments and system-level trade-offs are performed to identify the most suitable mission approach.

Phase A – Feasibility and Conceptual Design

The feasibility of LAZARUS is assessed during this phase. Different system architectures and technical solutions are analysed and compared. Preliminary estimates of spacecraft performance, mass, power, propulsion requirements, and other key system parameters are established. The phase concludes with the selection of the baseline mission concept.

Phase B – Preliminary Design

The selected mission concept is developed into a preliminary spacecraft design. The overall spacecraft architecture is defined, and the main subsystems are designed at a preliminary level. Interfaces between subsystems are established, and the main system budgets are defined and consolidated.

Phase C – Detailed Design

The preliminary design is further developed into a detailed design suitable for manufacturing, assembly, integration, and testing. Detailed analyses are performed for all spacecraft subsystems, and the required components and materials are specified. Components that cannot be manufactured in-house are procured from external suppliers.

Phase D – Manufacturing, Assembly, Integration and Testing

The spacecraft components are manufactured in-house or procured from external suppliers. The components are subsequently assembled and integrated by astroFUEL. The integrated spacecraft is subjected to the required verification and qualification tests to demonstrate compliance with the applicable system and subsystem requirements and to verify its readiness for launch.

Phase E – Launch and Operations

The spacecraft is launched and commissioned in orbit. Following successful commissioning, astroFUEL conducts the planned mission operations, including spacecraft control, rendezvous and proximity operations, and the refuelling of client satellites (only as a technology demonstrator).

Phase F – End of Life

At the end of its operational lifetime, the spacecraft is removed from active service and the appropriate EOL and disposal procedures are implemented in accordance with the applicable mission requirements and regulatory constraints.

5 Project planning

5.1 Project planning of the Preliminary Design Project

This section details the strategic and operational planning for the preliminary design of the LAZARUS mission, covering the activities corresponding to Phases 0, A, and B. To ensure efficient execution, resource management has been structured based on the work packages previously defined in the Work Breakdown Structure (WBS).The Responsibility Assignment Matrix (RAM) documents the distribution of tasks among the technical areas using a standardized nomenclature. The R (Responsible) role identifies the main department in charge of leading. Meanwhile, the S (Support) role designates the areas acting as support.

Furthermore, this section establishes the logical diagram and the project's schedule. These schemas sequence the parallel development of the various subsystems through to their integration, culminating in the primary milestone that marks the conclusion of this stage: the Preliminary Design Release (PDR). This review will consolidate the satellite's baseline architecture and establish the critical path that will define the preliminary project.

Work Package	**PM**	**MA**	**M&S**	**Prop**	**AOCS/ GNC**	**Pwr**	**Thrm**	**C&D**	**Pay**
WP-1000	**R**								
WP-1100	**R**								
WP-1110	S		R						
WP-1110.1	S		R						S
WP-1110.2	S		R						
WP-1110.3	S		R						S
WP-1110.4	S		R	S					
WP-1110.5	S	S	R	S	S	S	S	S	S
WP-1110.6	S		R			S	S	S	S
WP-1120	S				R				
WP-1120.1	S	S			R			S	
WP-1120.2	S			S	R				
WP-1120.3	S	S			R			S	S
WP-1120.4	S		S		R	S			
WP-1130	S			R					
WP-1130.1	S	S		R					
WP-1130.2	S	S		R					
WP-1130.3	S		S	R					
WP-1130.4	S	S		R					
WP-1130.5	S		S	R	S				
WP-1130.6	S		S	R					S
WP-1140	S							R	
WP-1140.1	S	S						R	

Work Package	PM	MA	M&S	Prop	AOCS/ GNC	Pwr	Thrm	C&D	Pay
WP-1140.2	S				S			R	S
WP-1140.3	S							R	
WP-1140.4	S							R	
WP-1150	S					R			
WP-1150.1	S	S	S	S	S	R	S	S	S
WP-1150.2	S				S	R		S	
WP-1150.3	S		S			R			
WP-1150.4	S					R	S		
WP-1150.5	S					R		S	
WP-1160	S						R		
WP-1160.1	S	S	S	S	S	S	R	S	S
WP-1160.2	S	S					R		
WP-1160.3	S					S	R		
WP-1160.4	S		S				R		
WP-1200	**S**								**R**
WP-1210	S								R
WP-1210.1	S		S						R
WP-1220	S								R
WP-1220.1	S			S					R
WP-1220.2	S						S		R
WP-1230	S								R
WP-1230.1	S	S			S			S	R
WP-1300	**R**	**S**							
WP-1320	R	S							
WP-1320.1	R	S							
WP-1320.2	R	S		S					
WP-1330	R	S							
WP-1330.1	R	S			S				
WP-1340	R	S							
WP-1340.1	R	S			S				S
WP-1350	R	S							
WP-1350.1	R	S	S	S					
WP-3000	**R**	**S**	**S**						
WP-3100	R	S	S						
WP-3100.1	R	S	S						
WP-3100.2	R	S	S						
WP-3300	R	S							
WP-3300.1	R	S							

Table 5.1: Responsibility Assignment Matrix.

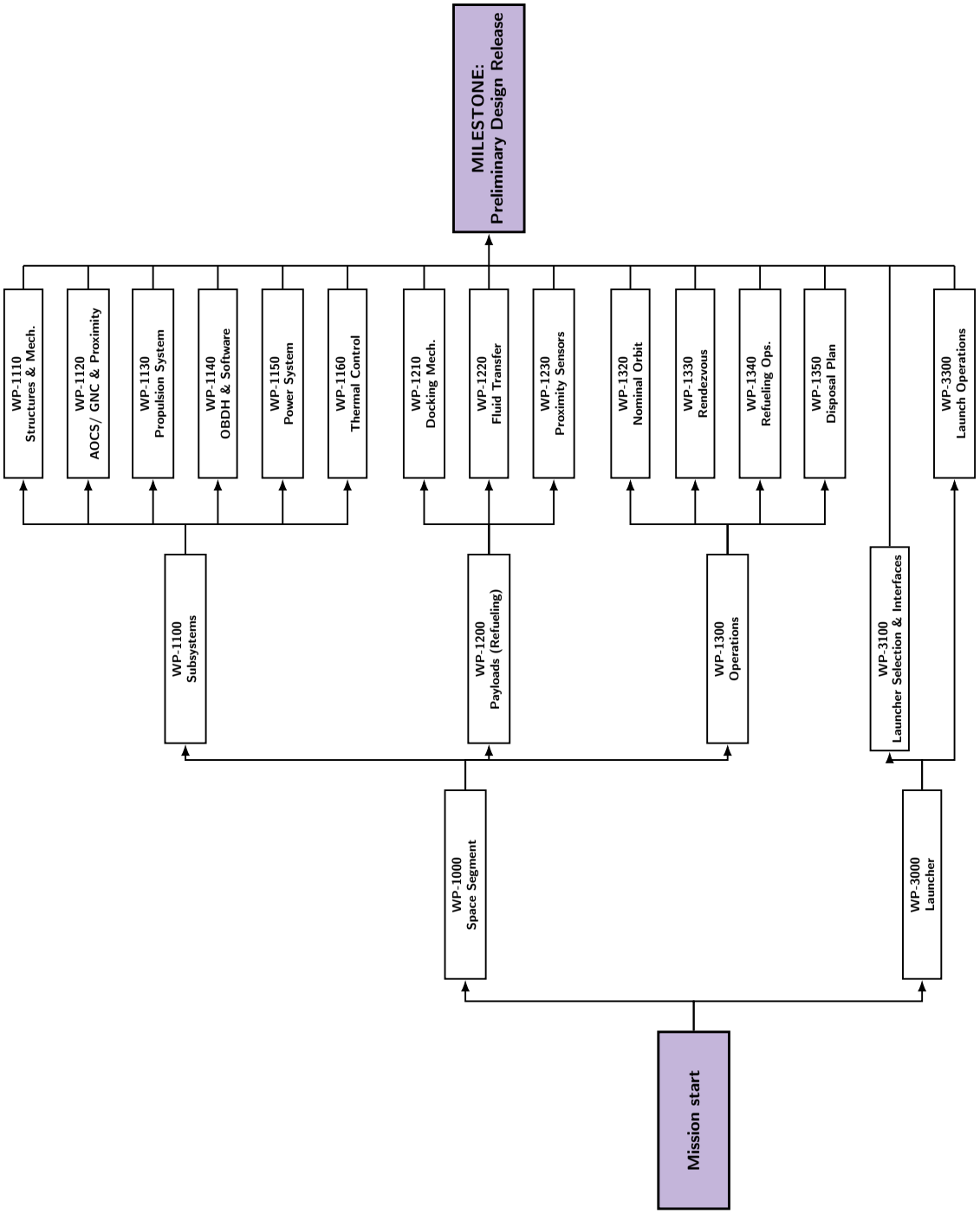


Figure 5.1: WBS logical diagram (Phases 0, A, and B)

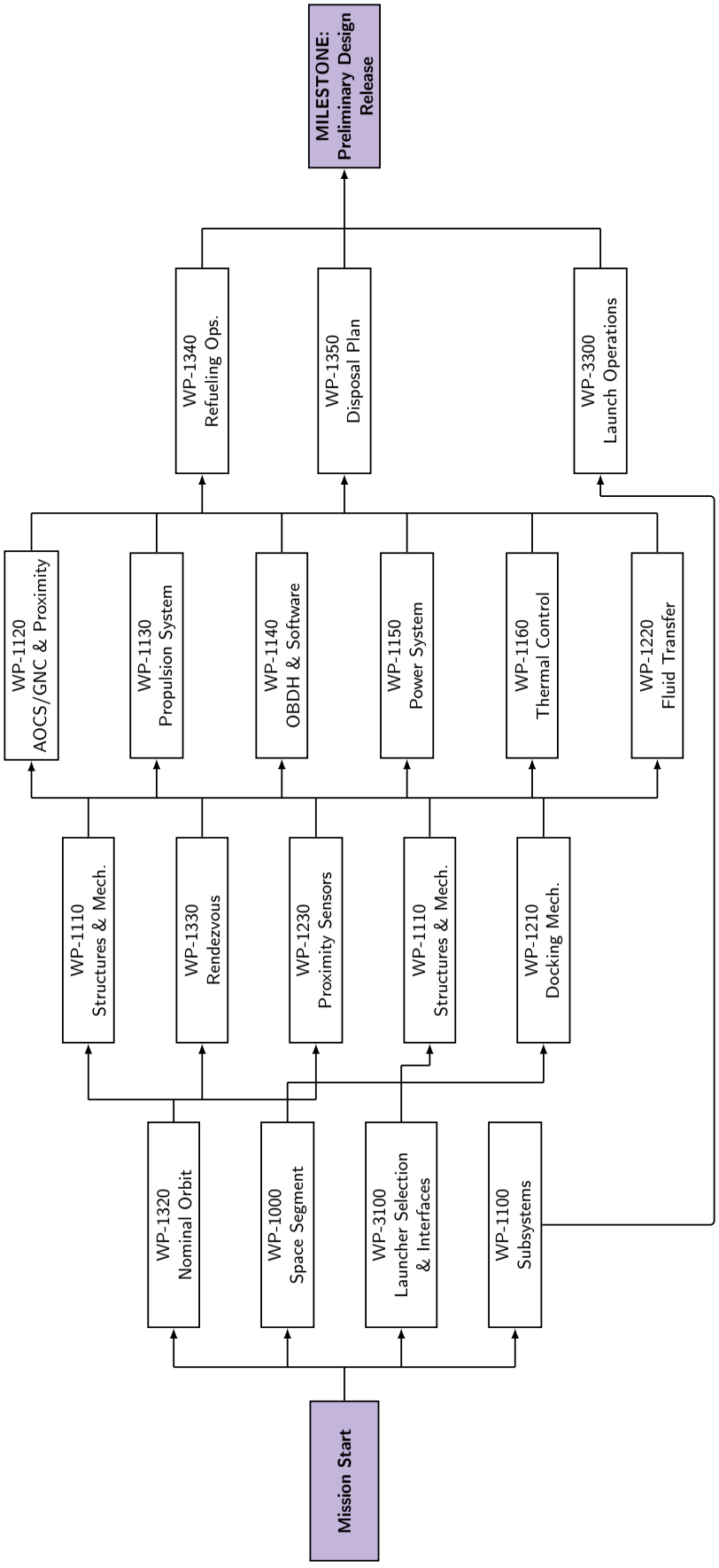


Figure 5.2: Figure 5.2: Precedence Diagram

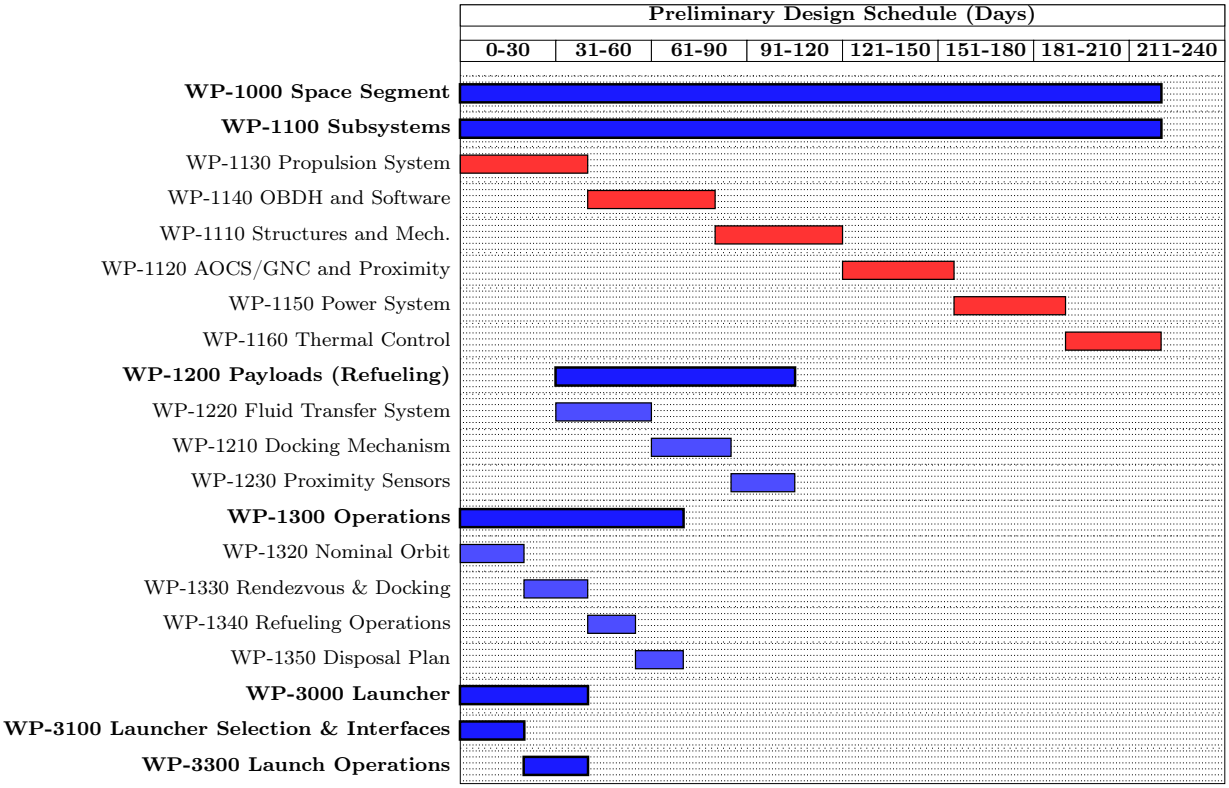


Figure 5.3: Preliminary Design Schedule (Phases 0, A, and B). The Critical Path through the subsystems is highlighted in red.

5.2 Project planning of the complete Project

This section extends the planning of the preliminary design to the complete LAZARUS lifecycle, from the initial mission analysis (Phase 0) to the End of Life (Phase F). It follows the same structure as the previous section: the distribution of responsibilities among the technical areas is first defined through the Responsibility Assignment Matrix, then the logical sequence of the work packages is established, and finally the schedule of the whole project is presented, culminating in the Mission Closure milestone.

The Responsibility Assignment Matrix of the complete project extends the preliminary RAM to the full LAZARUS lifecycle (Phases 0+A to F), following the WBS defined in Section 7.2. The same nomenclature is used: "R" (Responsible) identifies the area leading each work package, while "S" (Support) designates the areas contributing to it. Management, launcher and operations coordination remain under the Project Manager, while each Level 4 technical activity is led by the area owning the corresponding subsystem or domain.

Work Package	PM	MA	M&S	Prop	AOCS/ GNC	Pwr	Thrm	C&D	Pay
WP-0000	**R**								
WP-0100	**R**								
WP-0110	R								
WP-0120	R	S							
WP-0130	R								
WP-0140	R	S	S	S	S	S	S	S	S
WP-0200	**R**								
WP-0210	R	S	S	S	S	S	S	S	S
WP-0220	R	S	S	S	S	S	S	S	S
WP-0230	R	S	S	S	S	S	S	S	S
WP-0240	R								
WP-0300	**R**								
WP-0310	R								
WP-0320	R								
WP-0330	R								
WP-0400	**R**								
WP-0410	R								
WP-0420	R	S							
WP-0430	R	S							
WP-0440	R	S	S	S	S	S	S	S	S
WP-0450	R	S	S	S	S	S	S	S	S
WP-0460	R	S	S	S	S	S	S	S	S
WP-0470	R	S	S	S	S	S	S	S	S
WP-0480	R	S	S	S	S	S	S	S	S
WP-1000	**R**								

Work Package	PM	MA	M&S	Prop	AOCS/ GNC	Pwr	Thrm	C&D	Pay
WP-1100	**R**								
WP-1110	S		R						
WP-1111	S	S	R	S					S
WP-1112	S		R				S		
WP-1113	S		R						
WP-1114	S		R	S	S	S	S	S	S
WP-1120	S				R				
WP-1121	S	S	S		R	S			
WP-1122	S	S		S	R			S	
WP-1123	S				R			S	S
WP-1124	S			S	R			S	S
WP-1130	S			R					
WP-1131	S	S	S	R					
WP-1132	S		S	R			S		
WP-1133	S		S	R					S
WP-1134	S			R	S		S		
WP-1140	S							R	
WP-1141	S					S		R	
WP-1142	S				S			R	S
WP-1143	S				S	S		R	S
WP-1144	S				S			R	
WP-1150	S					R			
WP-1151	S	S	S			R			
WP-1152	S	S				R	S		
WP-1153	S		S			R		S	
WP-1154	S		S	S	S	R		S	S
WP-1160	S						R		
WP-1161	S		S				R		
WP-1162	S					S	R	S	
WP-1163	S	S		S		S	R		S
WP-1164	S		S			S	R	S	S
WP-1200	**S**								**R**
WP-1210	S								R
WP-1211	S		S						R
WP-1212	S		S		S	S			R
WP-1213	S		S		S				R
WP-1220	S								R
WP-1221	S			S		S			R
WP-1222	S		S	S			S		R
WP-1223	S			S				S	R

Work Package	PM	MA	M&S	Prop	AOCS/ GNC	Pwr	Thrm	C&D	Pay
WP-1230	S								R
WP-1231	S				S			S	R
WP-1232	S				S	S			R
WP-1233	S				S			S	R
WP-1300	**R**	**S**							
WP-1310	S		R						
WP-1311	S		R	S	S	S	S	S	S
WP-1312	S		R	S	S	S	S	S	S
WP-1313	S		R			S	S	S	
WP-1320	R	S							
WP-1321	R	S		S	S	S		S	
WP-1322	R	S	S	S	S	S	S	S	S
WP-1323	R	S				S	S	S	
WP-1330	R	S							
WP-1331	S	R		S	S				
WP-1332	S				R			S	S
WP-1333	S		S		S				R
WP-1340	R	S							
WP-1341	S		S						R
WP-1342	S			S			S	S	R
WP-1343	S				S				R
WP-1350	R	S							
WP-1351	R			S		S			S
WP-1352	S	R		S	S				
WP-1353	R							S	
WP-2000	**R**							**S**	
WP-2100	**S**							**R**	
WP-2110	S	S						R	
WP-2120	R							S	
WP-2130	S				S			R	S
WP-2200	**S**							**R**	
WP-2210	S				S	S	S	R	
WP-2220	S			S				R	S
WP-2230	R	S			S			S	S
WP-3000	**R**	**S**	**S**						
WP-3100	**R**	**S**	**S**						
WP-3110	R	S	S						
WP-3120	R	S	S	S					
WP-3130	R								
WP-3140	S		R						

Work Package	PM	MA	M&S	Prop	AOCS/ GNC	Pwr	Thrm	C&D	Pay
WP-3150	S		R						
WP-3160	S		S			R		S	
WP-3200	**R**		**S**						
WP-3210	S		R						
WP-3220	S		R			S		S	
WP-3230	S			R					S
WP-3300	**R**	**S**							
WP-3310	R					S		S	
WP-3320	S	R						S	
WP-3330	R	S	S		S				

Table 5.2: Responsibility Assignment Matrix of the complete project.

The planning of the complete project is presented through the WBS logical diagram (Figure 5.4), the precedence diagram (Figure 5.5) and a set of schedules based on the WBS of Section 7.2. The complete project spans 68 months: Phases 0+A (3 months) and B (5 months) correspond to the 240-day preliminary design described in the previous section, followed by Phase C (12 months), Phase D (18 months), Phase E (24 months) and Phase F (6 months). The schedules are organised by WBS branch (Management, Subsystems, Payloads and Operations, and Ground Segment and Launcher), all on the same 68-month time scale, so that each work package appears complete in a single figure. Monitoring and control tasks (0130, 0240, 0320, 0330 and 0410) are continuous level-of-effort activities spanning the whole lifecycle. The detailed day-level planning of Phases 0+A and B is given in the preliminary design schedule of the previous section.

Figure 5.4 shows how the four Level 1 branches of the WBS (Management, Space Segment, Ground Segment and Launcher) start in parallel from the beginning of the mission and converge on the Mission Closure milestone. Figure 5.5 refines this view by establishing the dependencies between work packages. The project is divided into two parts: a development part (Phases 0+A to D), in which the subsystems, payload, ground segment and launcher interfaces are developed in parallel and converge on the integration and testing of the spacecraft and the Flight Readiness Review (FRR); and an operational part (Phases D to F), in which launch, commissioning, rendezvous, refuelling and disposal take place strictly in sequence, since each activity cannot start until the previous one is completed. The management activities run continuously in parallel with both parts until the end of the mission.

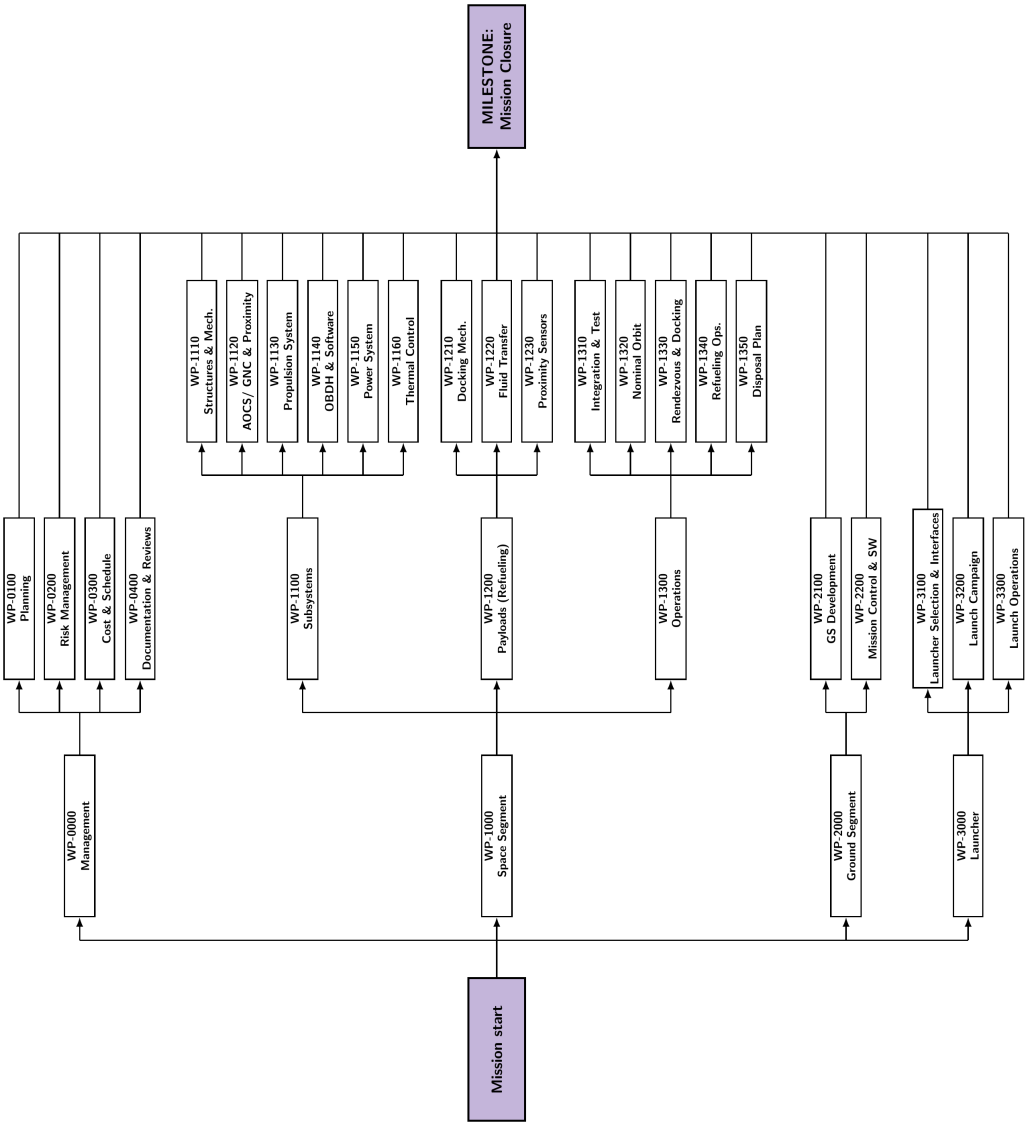


Figure 5.4: WBS logical diagram of the complete project (Phases 0+A to F).

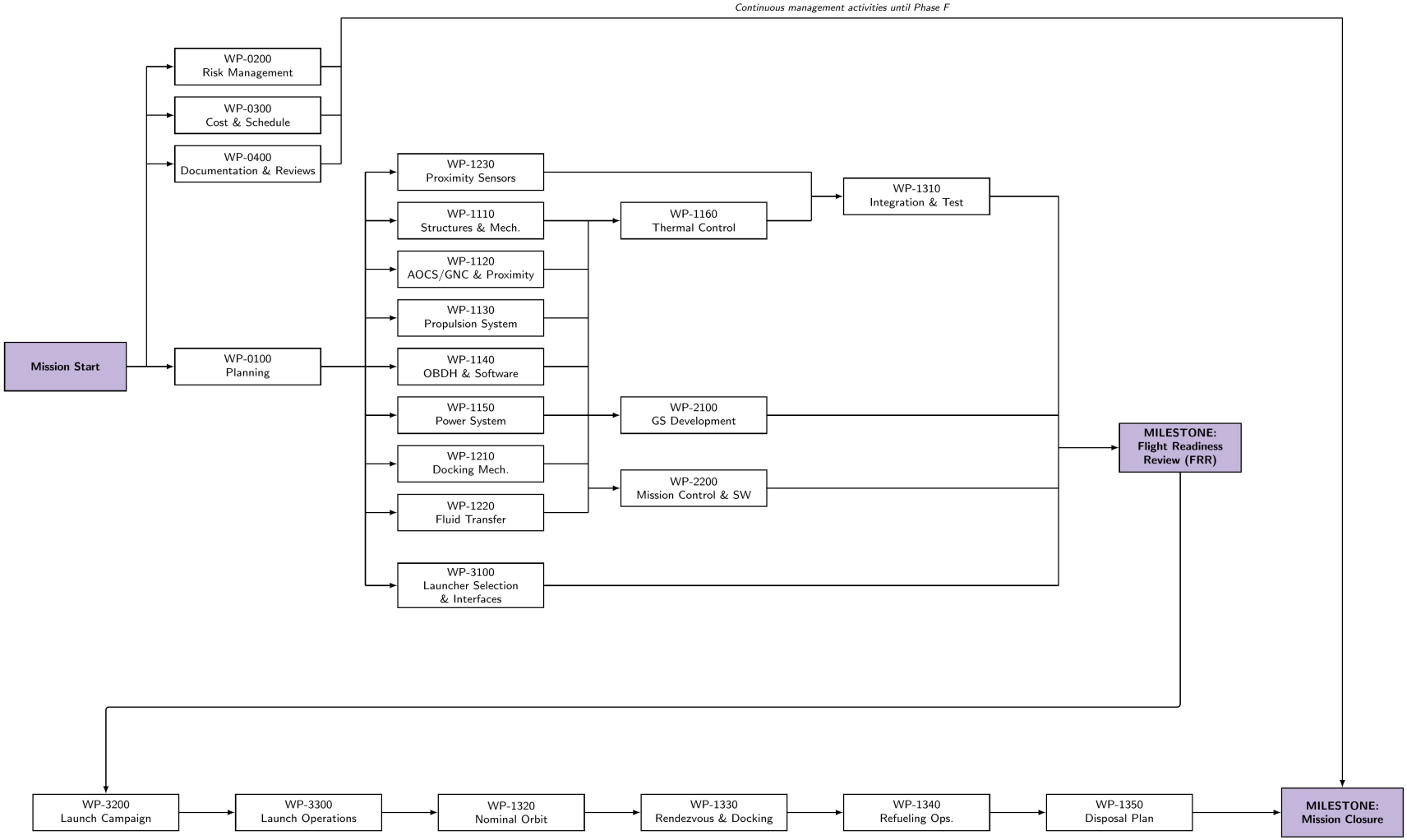


Figure 5.5: Precedence diagram of the complete project (Phases 0+A to F).

The main milestones of the complete project are the following: the Preliminary Design Review (PDR) closes the preliminary design at month 8; the Critical Design Review (CDR) validates the detailed design at months 19–20; the Qualification Request (QR) and the Flight Acceptance Review (FAR) confirm, at months 35–36, that the integrated spacecraft is qualified and ready to be shipped; the launch takes place at months 38–40; the refuelling demonstration on the client satellite is carried out at months 51–56; and the End of Mission is declared at month 68, after the spacecraft has been passivated and moved to the graveyard orbit.

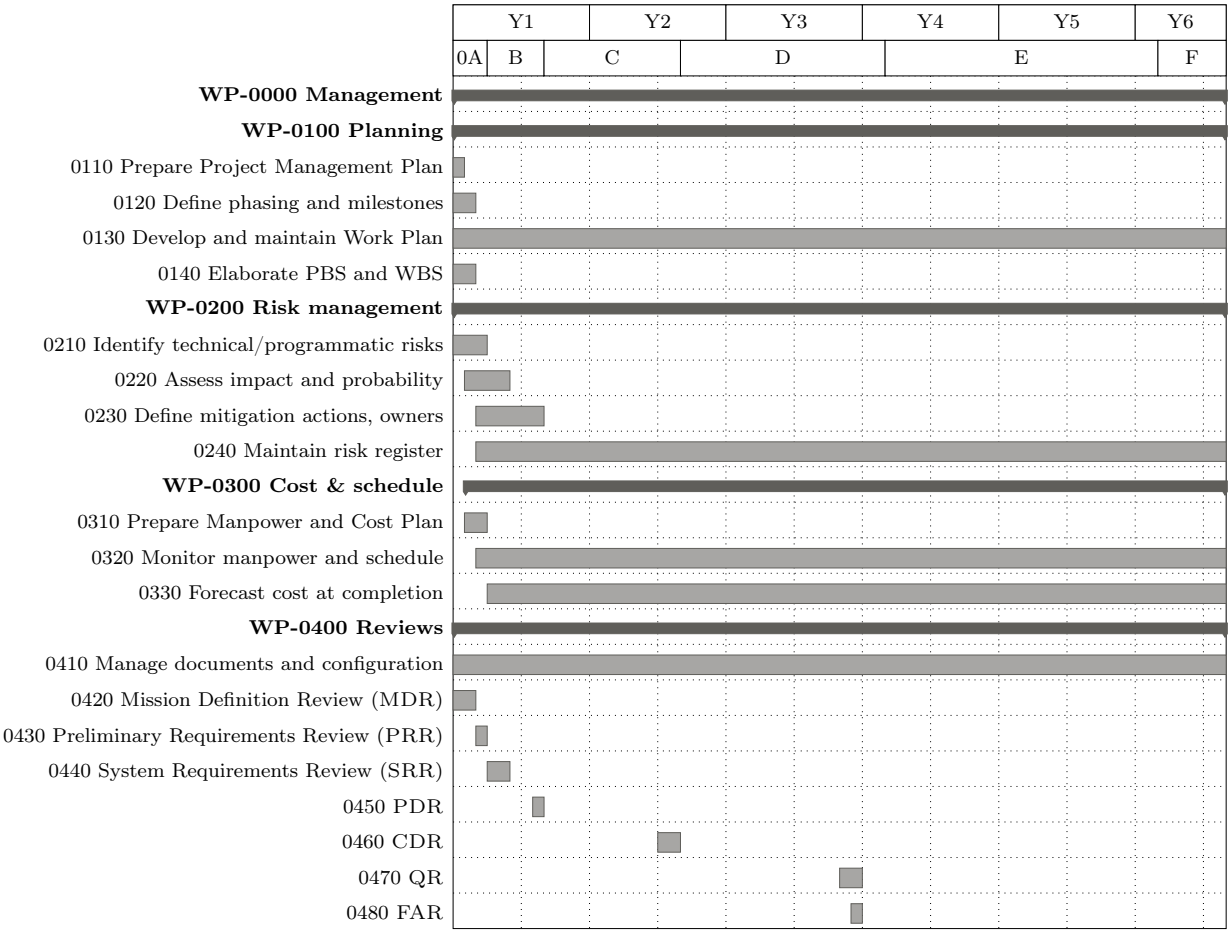


Figure 5.6: Schedule of WP-0000 Management over the complete project (Level 4 tasks). Time is in project months (1 month = 30 days).



Figure 5.7: Schedule of WP-1100 Subsystems over the complete project (Level 4 tasks). Time is in project months.

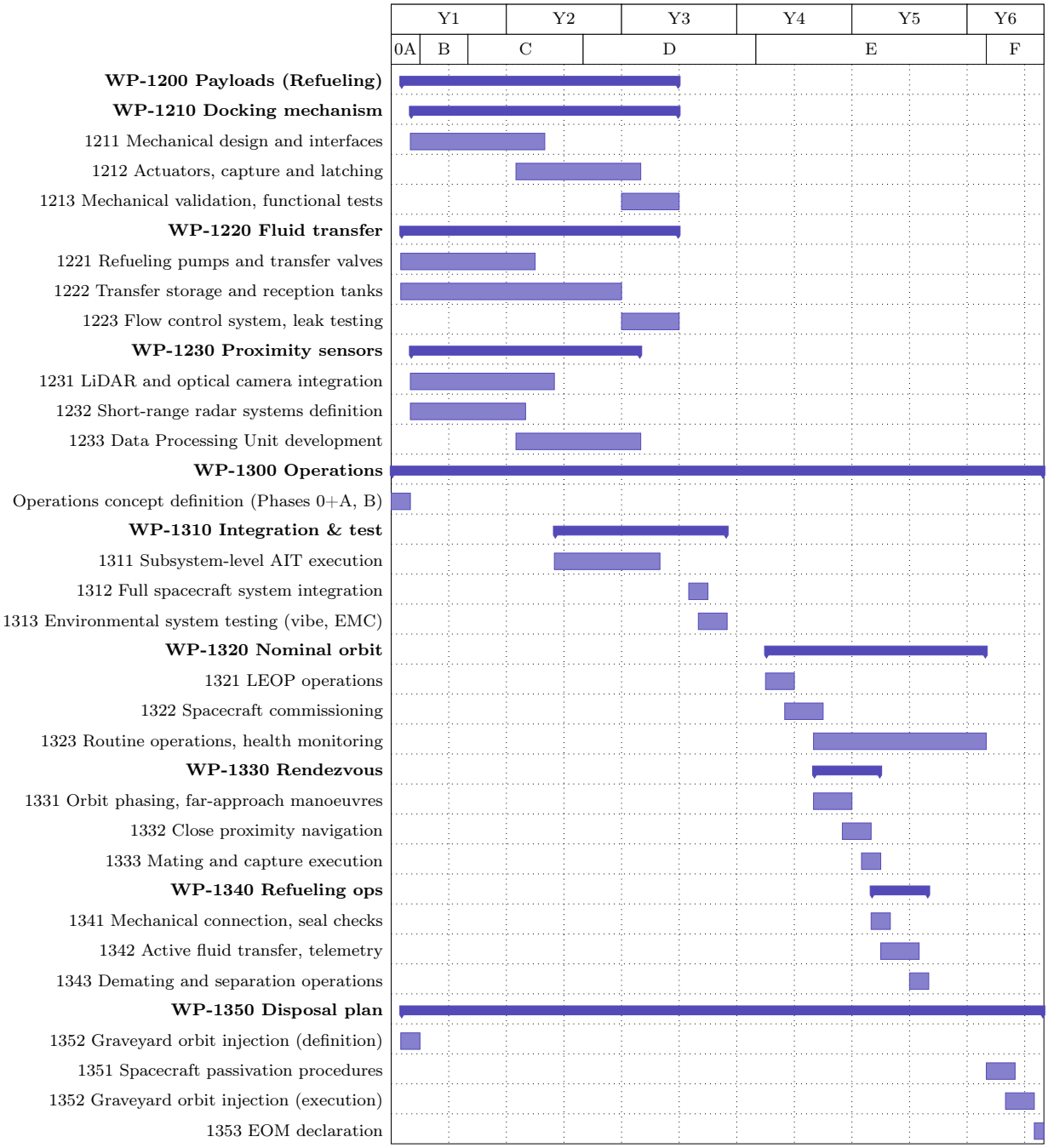


Figure 5.8: Schedule of WP-1200 Payloads and WP-1300 Operations over the complete project (Level 4 tasks). Time is in project months.



Figure 5.9: Schedule of WP-2000 Ground Segment and WP-3000 Launcher over the complete project (Level 4 tasks). Time is in project months.

The critical path of the complete project runs through the last subsystem qualification activities (hardware-in-the-loop testing, propulsion firing tests, software validation and thermal vacuum testing, WP-1124, WP-1134, WP-1144 and WP-1164), the full spacecraft integration and environmental testing (WP-1312 and WP-1313), the QR and the FAR, the launch campaign and launch operations (WP-3200 and WP-3300), and the in-orbit sequence of commissioning, rendezvous and refuelling (WP-1320 to WP-1340). These activities are strictly chained, since each of them requires the previous one to be completed, so any delay among them directly postpones the refuelling of the client satellite, which, as identified in Section 6, could lead to a breach of the service contracts. The remaining work packages have float with respect to this path: the launcher interfaces (WP-3130 to WP-3160) are completed about twenty months before the launch campaign, the Ground Segment is validated four months before launch, and several subsystem and payload activities finish ahead of the spacecraft integration. Since the Phase D activities are tightly chained, schedule margins should be negotiated in the launch slot and in the time available between the refuelling demonstration (month 56) and the end of routine operations (month 62), which allows a delay of up to six months to be absorbed without affecting the End of Mission.

6 Risk Management

6.1 Identification of main risks

The LAZARUS mission, part of the AstroFUEL initiative, focuses on developing an orbital refuelling infrastructure to extend the life of satellites in geostationary orbit (GEO). Its most critical segment is the refuelling maneuver itself, which requires highly accurate and coordinated work between the propulsion and payload subsystems to provide propellant to fully functional satellites that simply lack the fuel to maintain their orbit. Being a trailblazer mission for this type of operation, it needs to prove its reliability to present and future clients while actively working to reduce space debris. Due to this demanding nature, the project carries a series of risks that must be carefully evaluated, managed, and limited.

1. **Not accurate docking maneuver**
   LAZARUS must perform delicate rendezvous and docking maneuvers to correctly approach the client satellite. The satellite must remain securely attached through its standard refuelling interface for the entire duration of the refuelling process, avoiding any unintended perturbation of the client's orbit. The biggest risk during this proximity operation involves interfering with the client's mission. High accuracy is required to prevent a worst-case scenario: a collision between the two bodies, which would result in the absolute failure of the mission and the creation of new space debris, contradicting the mission's primary goal. Furthermore, the docking mechanism must ensure a safe separation once the transfer is complete.

2. **Malfunctioning during the refueling**
   The LAZARUS satellite carries the necessary propellant stored within high-pressure tanks. During the transfer phase, the fuel pump, the pressurization system, and the standard refuelling interface must work perfectly. A malfunction or a leak during this highly critical phase could alter the specific conservation conditions of the Xenon required inside the client's tank, rendering the entire refuelling operation useless and wasting valuable resources.

3. **Fuel waste and Delta-V management**
   The Mission Specialist is responsible for defining the main orbit parameters and minimizing the Delta-V requirement across all mission phases, including launch, transfer, operational, and de-orbit. LAZARUS must complete a series of complex maneuvers to reach the orbits of multiple different clients and must be ready to repeat the process throughout its lifetime. However, unexpected events, such as maneuvers required to avoid space debris, could demand extraordinary propellant consumption. If the Delta-V is not perfectly optimized, this could excessively deplete the satellite's propellant reserves, leaving it without enough fuel to reach the next client or to safely enter a graveyard orbit at the end of its mission.

4. **Component performance and lack of flight heritage**
   Due to the unprecedented nature of this mission, there is an inherent risk that newly developed components may not perform in the space environment exactly as they did during tests on Earth, presenting typical issues associated with a lack of flight heritage. Specifically, the Mechanical Specialist must define highly reliable mechanisms for all elements that need to be deployed or pointed. Any anomaly in these mechanisms, such as a failure in the innovative docking mechanism, could completely prevent the satellite from interacting with the client.

5. **Schedule Risks**
   The AstroFUEL project operates under commercial contracts intended for the 2030s market, which are often stipulated prior to the launch of the client satellites. Therefore, massive delays during the development and launch phases of LAZARUS are unacceptable. If the schedule slips significantly, LAZARUS might not be able to reach and refuel the client in time to successfully extend its operational life, leading to a breach of contract and severe economic repercussions for the project.

6. **Market and economics risks**
   While the LAZARUS mission operates on contracts stipulated prior to launch, it remains highly exposed to broader market and economic threats. The primary concern is the extreme price volatility and limited global supply of Xenon gas required to fill the satellite's high-pressure tanks. A sudden spike in raw material costs could drastically increase the project's baseline budget, compromising the mission's financial viability. Furthermore, there is the risk that a pre-contracted client satellite might suffer an independent critical failure before LAZARUS can reach it. If a client is lost, LAZARUS must quickly secure a replacement target in orbit to sell its payload, or risk finishing the mission with unsold propellant, thereby nullifying the expected economic return of the AstroFUEL initiative.

6.2 Characterization of main risks in terms of impact and probability

Once the primary risks are identified, it is necessary to characterize them by evaluating their likelihood of occurrence and the severity of their consequences on the mission's performance requirements. This evaluation determines the overall criticality of each risk and guides the development of targeted mitigation strategies.

1. **Not accurate docking maneuver**

   The autonomous proximity operations in geostationary orbit are subjected to sensor noise, lighting variations and target dynamics that may not be accurately predicted since they have

a level of uncertainty intrinsic in the maneuver, while with a medium probability level. A collision of the satellite with the target in this phase of the mission carries a high level of risk for safety and technical aspects that varies from the loss of the mission to the generation of space debris, this aspect is crucial since the possible effect that can have also on the client's satellite causing a huge direct damage. To mitigate this risk the spacecraft must be equipped with a redundant proximity sensor, conveniently tested in the development phase, and an autonomous collision-avoidance software capable of executing an immediate abort sequence if during the maneuver predetermined tolerances are surpassed. The team will also dedicate resources to investigate similar errors and solutions applied in previous rendezvous missions.

2. **Malfunctioning during the refueling**

The refuelling process is particularly exposed to the harsh space environment while having to make a transfer of Xenon at high pressure, causing a significant stress on valves, pumps and seals used, that can aggravate for each refuelling cycle. This risk has an an high impact on the technical and economic aspect of the mission; a leak or pump failure would render the primary payload useless, failing to meet meet the client requirements and compromising the refuel not only for the actual client satellite but also for the following ones leading to a huge economical loss from both AstroFuel and the clients. In order to mitigate the risk are required a series of severe ground tests (TVAC) on the pumping system in vacuum Chambers to simulate the Thermal cycles and proving the resistance of the apparate to recurrent stresses, alongside of that it's needed to add redundancy in term of isolation valves to prevent fuel loss and wear over time.

3. **Fuel waste and Delta-V management**

The increasingly crowded space environment makes the need for debris avoidance maneuvers highly probable over the satellite lifespan. On top of that the satellite has to complete a series of orbital maneuvers in order to approach and then dock with the client satellite that may require a higher DeltaV than expected; this factor can be critical especially for longer missions. This series of maneuvers will not destroy the satellite but will impact on the technical and economical aspect of the mission reducing the number of clients LAZARUS can service and not allowing it to complete the expected refuels, lowering the overall return. This risk can be with a proactive analysis of the trajectories and by loading the satellite with a calculated propellant margin specifically dedicated to unexpected maneuvers.

4. **Component performance and lack of flight heritage**

Because LAZARUS is a first-of-its-kind mission introducing novel operations like Xenon refuelling in geostationary orbit, several critical mechanisms and payload elements customly made for the mission inherently lack space flight heritage. The primary risk is that unproven components, particularly the complex fluid transfer interfaces, may not perform in

the space environment exactly as they did during tests on Earth, and be more susceptible than expected to degradation in the radiation environment compromising some function of the mission. To proactively manage this lack of heritage, the project management will enforce a rigorous technology maturation plan aimed at elevating the Technology Readiness Level (TRL) well before the Critical Design Review. The technical management layer will assign subject matter experts to investigate lessons learned from previous fluid-transfer experiments and will heavily fund early-stage environmental testing. By involving manufacturing teams immediately in the prototyping phase, the project ensures that any required design modifications are identified early, leaving ample schedule margins for verification and reducing the epistemic uncertainty before the final flight hardware is built.

5. Schedule Risks

The aerospace industry is frequently subjected to delays, the timeline of the project becomes less controllable the more parts of the satellite are assigned to outer companies for their development. Delays in the schedule have a high impact on the mission in terms of contractual obligations with the clients, that risk to not being satisfied in the established time leaving a timespan when the client satellite doesn't have fuel. This can lead not only to financial penalties, as could be written on the contract, due to an economic damage caused to our clients, but also to a loss of market trust for future operations. This risk can be managed by implementing correct schedule margins during the planning phase, according to the client's time-necessities, and adopting parallel development pathways for the most critical payload components.

6. Market and economics risks

The probability of extreme Xenon price fluctuations or supply chain bottlenecks during the procurement phase is considered Medium-High, while the sudden loss of a pre-contracted client satellite is evaluated as Low-Medium probability but High impact. Both scenarios present a severe threat to the Cost and Schedule execution domains, potentially turning a technically successful mission into a financial failure. To mitigate the commodity risk, the financial management team will secure long-term, fixed-price supply contracts for Xenon gas at the very inception of the project, hedging against future market inflation. To address the risk of losing a client, the engineering and business team will actively maintain a prioritized backup list of potential secondary clients in GEO that lack propellant. By allocating early funding to develop versatile trajectory optimization models, the Mission Specialist will be ready to quickly calculate new rendezvous maneuvers for these backup clients without exceeding the Delta-V budget, ensuring the mission remains profitable even if the original market scenario changes.

Table 6.1: Risk assessment and evaluation of the main project risks.

No.	Type	Impact	Prob.	Risk	Comment
1	Tech / Cost	0.95	0.05	**Medium-High**	Catastrophic damage on both satellites
2	Tech	0.8	0.20	**High**	Commercial failure
3	Cost	0.2	0.40	**Low**	High frequency of debris but low impact
4	Tech	0.5	0.4	**Low**	Uncertainty for developed technologies
5	Sched	0.4	0.2	**Very Low**	Not respecting contracts with clients
6	Cost	0.6	0.2	**Low**	Enterprise risk

7 WBS

7.1 WBS of the Preliminary Design Project

The Preliminary Design WBS defines the work required to develop LAZARUS up to the Preliminary Design Review, covering the activities associated with Phases 0, A and B. Its objective is to provide a structured decomposition of the work to be performed during the preliminary design stage, ensuring consistency with the project planning, responsibility assignment and schedule defined in Section 5.1.

The WBS is organized around the Space Segment (WP-1000) and Launcher (WP-3000). The Space Segment is further divided into Subsystems (WP-1100), Payloads (WP-1200) and Operations (WP-1300). Each of these branches is decomposed into the corresponding subsystem- and activity-level work packages, as represented in Figure 7.1.

At this stage, the WBS is limited to the work necessary to establish the preliminary spacecraft architecture, define the main subsystem and payload concepts, analyse the mission and operational aspects, and select a compatible launch solution. Activities belonging to later development phases are therefore outside the scope of this preliminary WBS and are included in the Complete Project WBS presented in Section 7.2.

Table 7.1: Detailed scope of the LAZARUS Preliminary Design, including specific phases.

WP	Scope	Phases
WP-1000 Space Segment		
WP-1100 Subsystems		
WP-1110 Structures and Mech.	**WP-1111:** Define the preliminary spacecraft geometry.	B
	WP-1112: Select the materials required for the preliminary structural design.	B
	WP-1113: Define the spacecraft volume and external envelope.	B
	WP-1114: Estimate the structural mass of the spacecraft.	B
	WP-1115: Define the structural requirements of the spacecraft.	0+A–B
	WP-1116: Define the preliminary layout and accommodation of the spacecraft equipment.	B
WP-1120 AOCS and Proximity	**WP-1121:** Select the sensors required for attitude, orbit and relative navigation.	B
	WP-1122: Select the actuators required for spacecraft attitude and orbit control.	B
	WP-1123: Analyse the control required during rendezvous and proximity operations.	B
	WP-1124: Perform the preliminary sizing of the AOCS subsystem.	B
WP-1130 Propulsion System	**WP-1131:** Identify the ΔV requirements derived from the mission analysis.	0+A
	WP-1132: Select the propellant and propulsion technology.	0+A–B
	WP-1133: Perform the preliminary sizing of the tanks and propulsion system.	B
	WP-1134: Estimate the propellant mass required by the mission.	B
	WP-1135: Define the preliminary thruster configuration and layout.	B

Continued on next page

WP	Scope	Phases
	WP-1136: Define the propellant storage system.	B
WP-1140 OBDH and Software	**WP-1141:** Define the ground-to-spacecraft communication links.	0+A
	WP-1142: Define the communications with the client spacecraft.	0+A
	WP-1143: Define the data handling and management functions.	B
	WP-1144: Develop the preliminary communications and data-handling architecture.	B
WP-1150 Power System	**WP-1151:** Estimate the spacecraft power consumption.	B
	WP-1152: Define the operational modes and their corresponding power demand.	B
	WP-1153: Perform the preliminary sizing of the solar arrays.	B
	WP-1154: Perform the preliminary sizing of the batteries.	B
	WP-1155: Define the preliminary electrical architecture.	B
WP-1160 Thermal Control	**WP-1161:** Identify the main thermal loads affecting the spacecraft.	B
	WP-1162: Define the operational thermal environment.	B
	WP-1163: Select the active and passive thermal control systems.	B
	WP-1164: Develop the preliminary thermal design.	B
WP-1200 Payload		
WP-1210 Docking Mechanism	**WP-1211:** Define the connection mechanisms and interfaces.	B
WP-1220 Fluid Transfer System	**WP-1221:** Define the propellant transfer requirements.	0+A–B
	WP-1222: Define the propellant transfer system.	B
WP-1230 Proximity Sensors	**WP-1231:** Define the docking system and procedure.	0+A-B

Continued on next page

WP	Scope	Phases
	WP-1300 Operations	
WP-1320 Nominal Orbit	**WP-1321:** Operational orbit selection.	0+A
	WP-1322: Analyse the transfers to the operational orbit.	0+A–B
WP-1330 Rendezvous and Docking	**WP-1331:** Analyse rendezvous and proximity operations.	0+A–B
WP-1340 Refueling Operations	**WP-1341:** Analyse the manoeuvres required for servicing operations in GEO.	0+A–B
WP-1350 Disposal Plan	**WP-1351:** Define the spacecraft EOL strategy.	0+A
WP-3000 Launcher		
WP-3100 Launcher Selection	**WP-3101:** Analyse the launch scenario.	0+A
	WP-3102: Select the launch vehicle.	0+A
WP-3300 Launch Operations	**WP-3301:** Define the injection orbit.	0+A–B

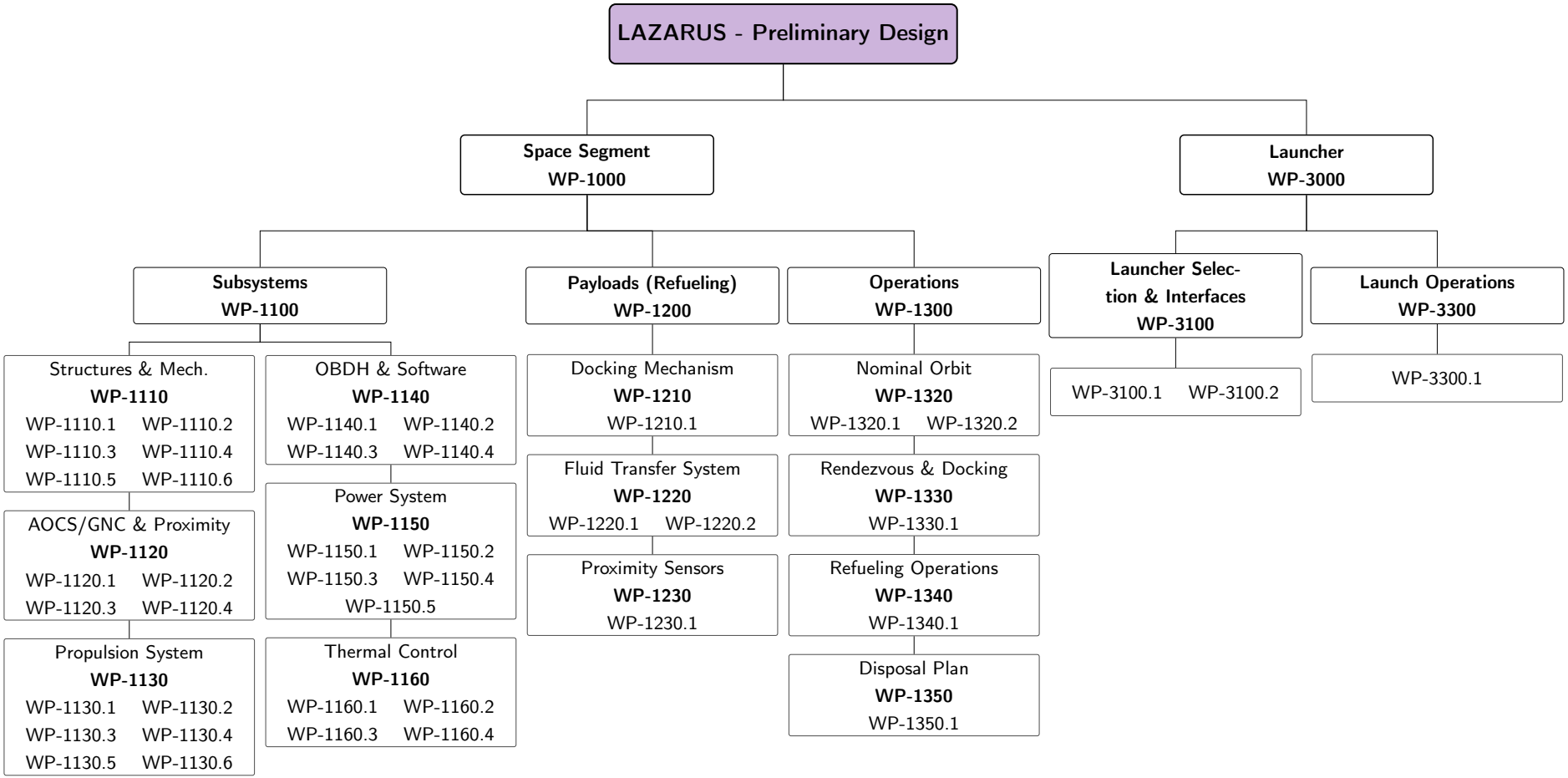


Figure 7.1: WBS for the LAZARUS Preliminary Design.

7.2 WBS of the complete Project

The LAZARUS WBS covers the entire mission lifecycle (Phases 0+A to F), establishing the baseline for planning, cost control, and risk management.

Structure and Numbering. The WBS uses a four-digit hierarchical code (thousands for Level 1, hundreds for Level 2, tens for Level 3, and units for Level 4):

- **Level 1:** Management (WP-0000), Space Segment (WP-1000), Ground Segment (WP-2000), and Launcher (WP-3000).
- **Levels 2 & 3:** Functional breakdowns (e.g., WP-1100 Platform, WP-1120 AOCS/GNC & Proximity).
- **Level 4:** Specific actionable packages (e.g., WP-1122 for GNC algorithm design).

This is shown in Figure 7.2, along branch details in Figures 7.3–7.9, and package scopes with European Cooperation for Space Standardization (ECSS) phases in Table 7.2.

Assumptions.

- The project scope extends through EOL disposal (Phase F, WP-1350).
- Subsystem equipment, whether developed in-house or procured, is integrated into its respective subsystem work package.
- Launcher selection & interfaces, campaign, and operations are consolidated entirely within the Launcher segment (WP-3000).

Table 7.2: Detailed scope of the LAZARUS WBS, including specific ECSS phases.

WP	Scope	Phases
WP-0000 Management		
WP-0100 Planning	**0110:** Prepare the Project Management Plan.	0+A–B
	0120: Define the project phasing and milestones.	0+A–B
	0130: Develop and maintain the Project Work Plan.	0+A–F
	0140: Elaborate the PBS and WBS.	0+A–B
WP-0200 Risk Mgmt	**0210:** Identify main technical and programmatic risks.	0+A–B
	0220: Assess impact and probability (Risk Matrix).	0+A–F
	0230: Define mitigation actions and assign owners.	0+A–F
	0240: Maintain risk register and follow up.	0+A–F
WP-0300 Cost & Sched.	**0310:** Prepare Manpower and Cost Plan.	0+A–B
	0320: Monitor manpower and schedule vs baseline.	0+A–F
	0330: Forecast cost at completion and report.	0+A–F
WP-0400 Reviews	**0410:** Manage documents and configuration.	0+A–F
	0420: Prepare and hold Mission Definition Review (MDR).	0+A
	0430: Prepare and hold PRR.	A
	0440: Prepare and hold SRR.	B
	0450: Prepare and hold PDR.	B
	0460: Prepare and hold CDR.	C
	0470: Prepare and hold QR.	D
	0480: Prepare and hold FAR.	D
WP-1000 Space Segment		
WP-1110 Structures	**1111:** Define requirements and trade-offs.	A–B
	1112: Detailed design and structural analysis, Finite Element Method (FEM).	B–C
	1113: Manufacturing of structural components.	C–D
	1114: Assembly, Integration and Testing (AIT).	D
WP-1120 AOCS/GNC & Prox	**1121:** Selection of sensors and actuators.	A–B
	1122: Design of GNC algorithms.	B–C
	1123: Development of the proximity vision system.	C–D
	1124: Hardware-in-the-Loop (HWIL) testing.	D

Continued on next page

WP	Scope	Phases
WP-1130 Propulsion	**1131:** Thrusters and propellant tanks sizing.	A–C
	1132: Feed system and valves detailed design.	B–C
	1133: Pressurization system integration.	C–D
	1134: Subsystem firing and qualification tests.	D
WP-1140 OBDH & SW	**1141:** On-Board Computer architecture and selection.	A–C
	1142: Data buses definition (CAN, SpaceWire).	B–C
	1143: Flight Software (FSW) development.	B–D
	1144: Software validation and V&V testing.	D
WP-1150 Power	**1151:** Solar arrays sizing and deployment mech.	A–C
	1152: Battery sizing and PCDU design.	B–C
	1153: Power distribution network layout.	C–D
	1154: Subsystem electrical integration.	D
WP-1160 Thermal	**1161:** Definition of Multi-layer insulation (MLI) and optical coatings.	B–C
	1162: Sizing of heaters and thermistors.	B–C
	1163: Thermal node analysis (Hot/Cold cases).	B–C
	1164: Thermal Vacuum test (TVAC).	D
WP-1210 Docking Mech	**1211:** Mechanical design and interfaces.	A–C
	1212: Actuators, capture and latching mechanisms.	C–D
	1213: Mechanical validation and functional testing.	D
WP-1220 Fluid Transfer	**1221:** Refueling pumps and transfer valves.	A–C
	1222: Transfer storage and reception tanks.	B–D
	1223: Flow control system and leak testing.	D
WP-1230 Prox. Sensors	**1231:** LiDAR and optical camera integration.	B–C
	1232: Short-range radar systems definition.	B–C
	1233: Data Processing Unit development.	C–D
WP-1310 Integ. & Test	**1311:** Subsystem-level AIT execution.	C–D
	1312: Full spacecraft system integration.	D
	1313: Environmental system testing (Vibe, EMC).	D
WP-1320 Nominal Orbit	**1321:** Launch and Early Orbit Phase (LEOP) operations.	E
	1322: Spacecraft commissioning.	E
	1323: Routine operations and health monitoring.	E

Continued on next page

WP	Scope	Phases
WP-1330 Rendezvous	**1331:** Orbit phasing and far-approach manoeuvres.	E
	1332: Close proximity navigation.	E
	1333: Mating and capture execution.	E
WP-1340 Refueling Ops	**1341:** Mechanical connection and seal checks.	E
	1342: Active fluid transfer and telemetry monitoring.	E
	1343: Demating and separation operations.	E
WP-1350 Disposal Plan	**1351:** Spacecraft passivation procedures.	F
	1352: Graveyard orbit injection manoeuvres.	B, F
	1353: End of Mission (EOM) declaration.	F
WP-2000 Ground Segment		
WP-2100 GS Dev.	**2110:** Ground Segment requirements and interfaces.	C
	2120: Procurement of stations and equipment.	C–D
	2130: Ground Segment validation and end-to-end tests.	D
WP-2200 Mission Ctrl	**2210:** Satellite control software (TM/TC).	C–E
	2220: Refueling specific telemetry dashboards.	C–E
	2230: Operational procedures and simulations.	C–D
WP-3000 Launcher		
WP-3100 Selection & Interfaces	**3110:** Identify launcher candidates.	A–B
	3120: Trade-off and mission compatibility.	A–B
	3130: Launch service contract preparation.	B–C
	3140: PLA interface definition and Interface Control Documents (ICD).	B–C
	3150: Separation system interface definition.	B–C
	3160: Umbilical interface definition.	B–C
WP-3200 Launch Camp.	**3210:** Shipping and handling to the launch site.	D
	3220: Launch site operations and integration.	D
	3230: Propellant loading and final checks.	D
WP-3300 Launch Ops	**3310:** Countdown and ascent phase.	D–E
	3320: Trajectory monitoring and flight tracking.	E
	3330: 3rd stage operations and spacecraft separation.	E

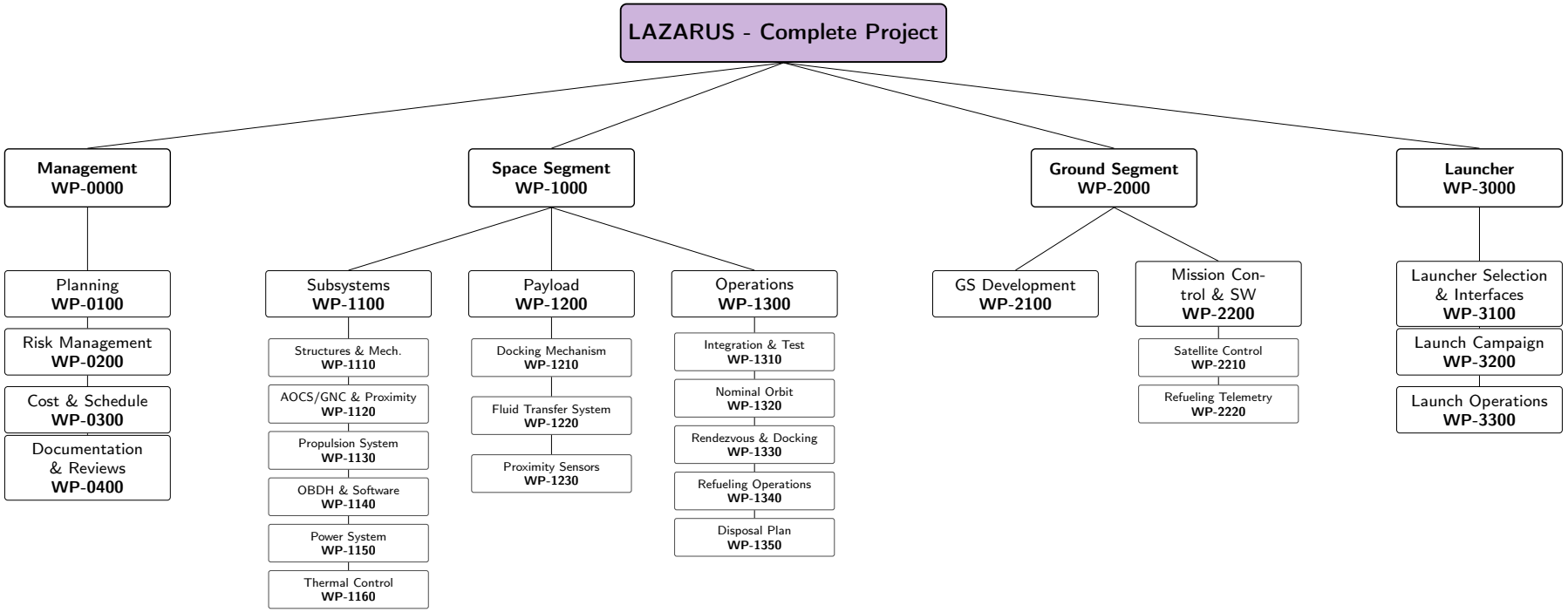


Figure 7.2: WBS for the LAZARUS Mission.

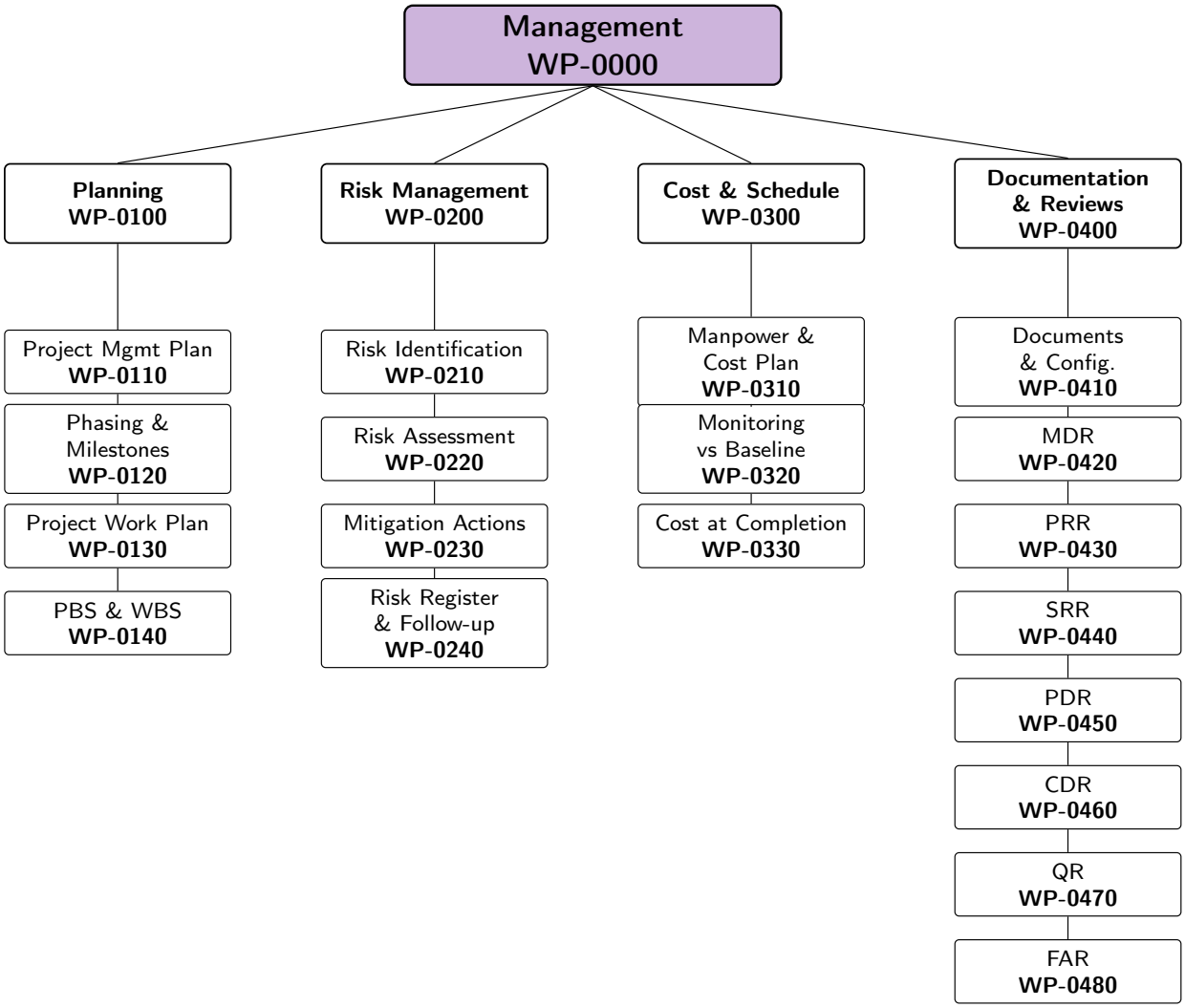


Figure 7.3: WBS of the Management branch (WP-0000).

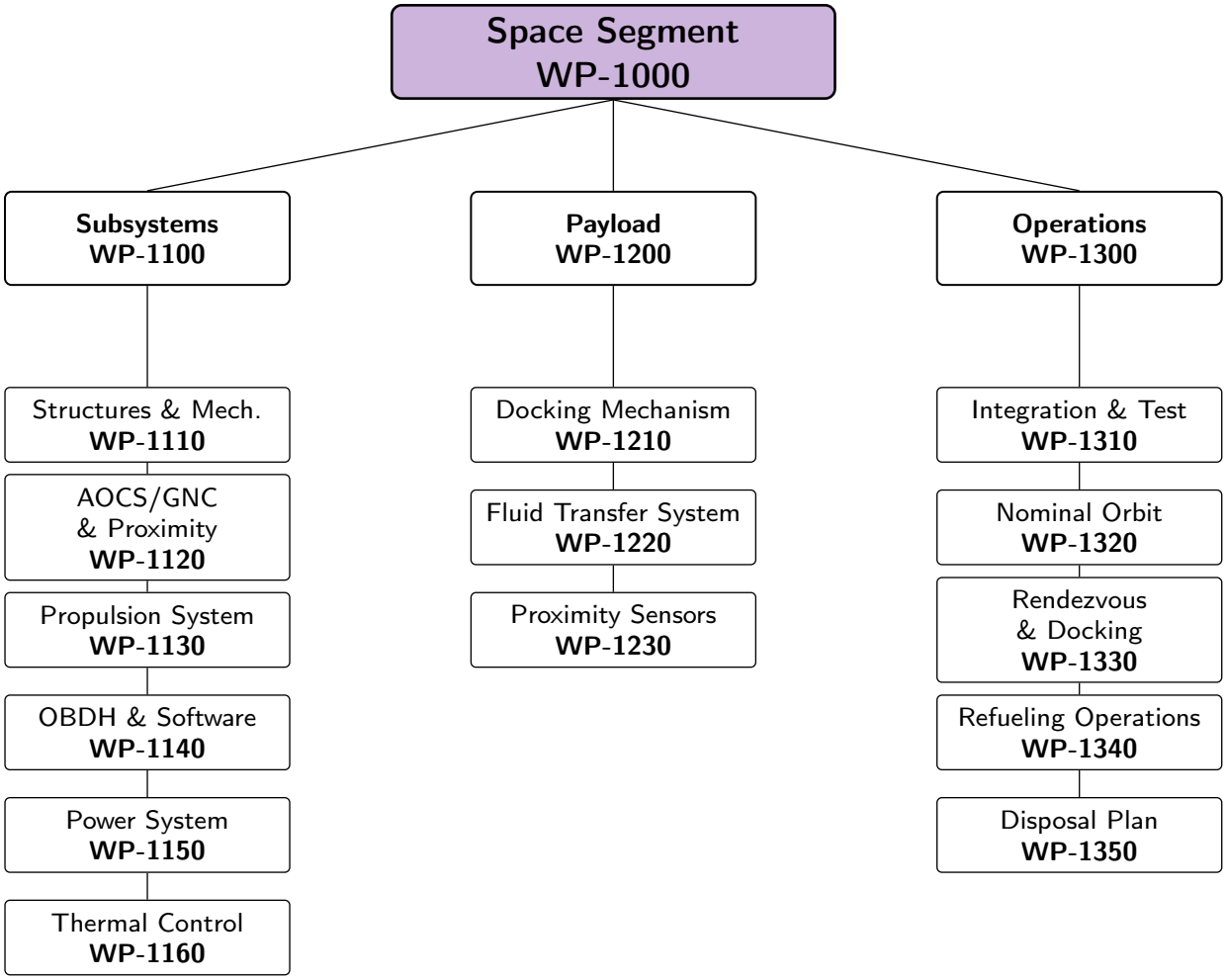


Figure 7.4: WBS of the Space Segment (WP-1000).

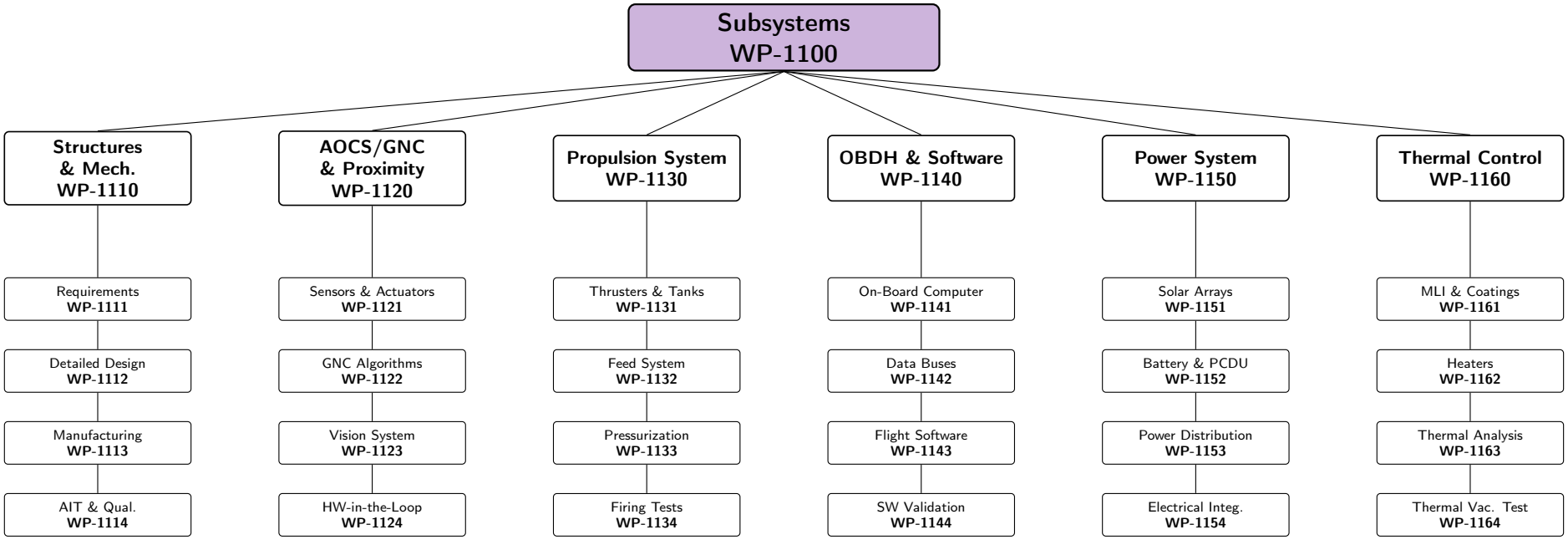


Figure 7.5: WBS of the Subsystems (WP-1100), down to Level 4.

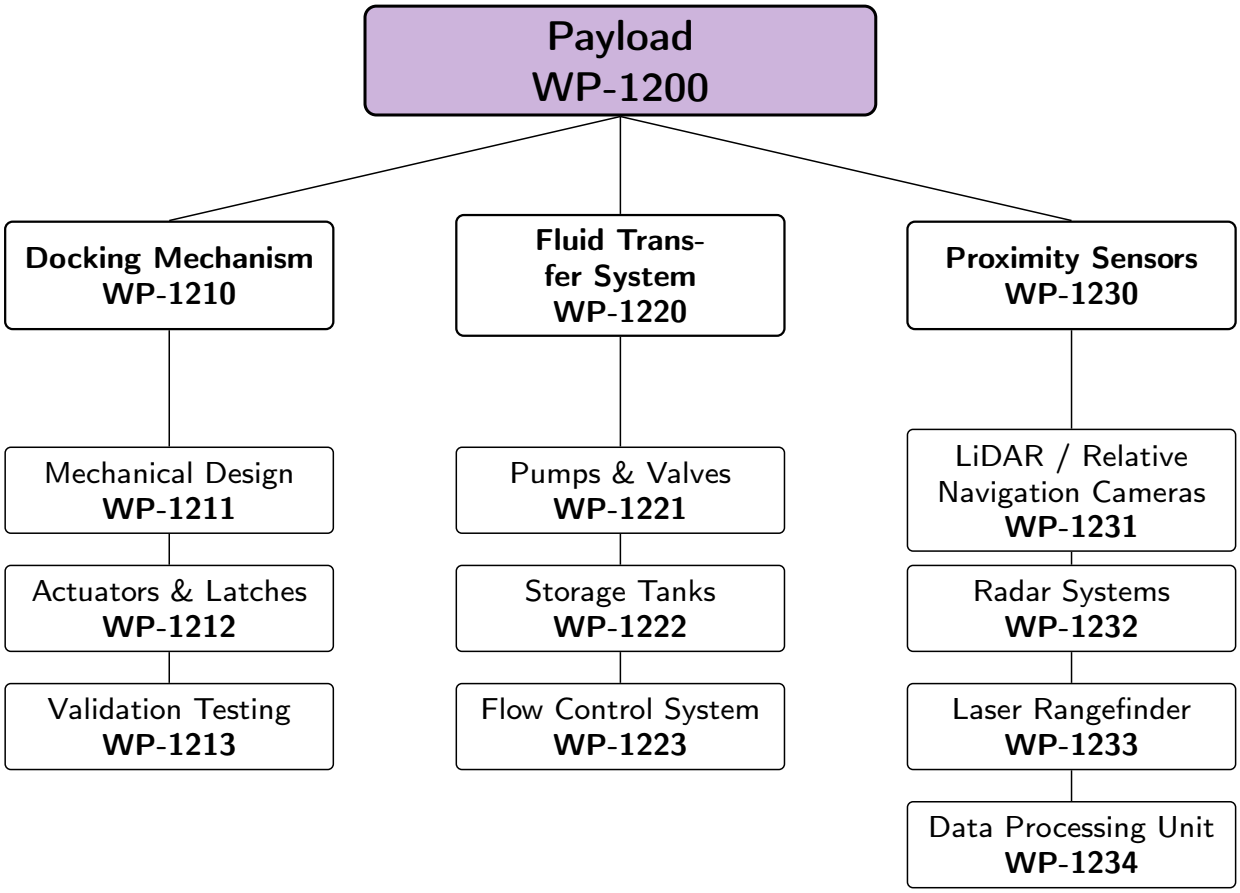


Figure 7.6: WBS of the Payload (WP-1200), down to Level 4.

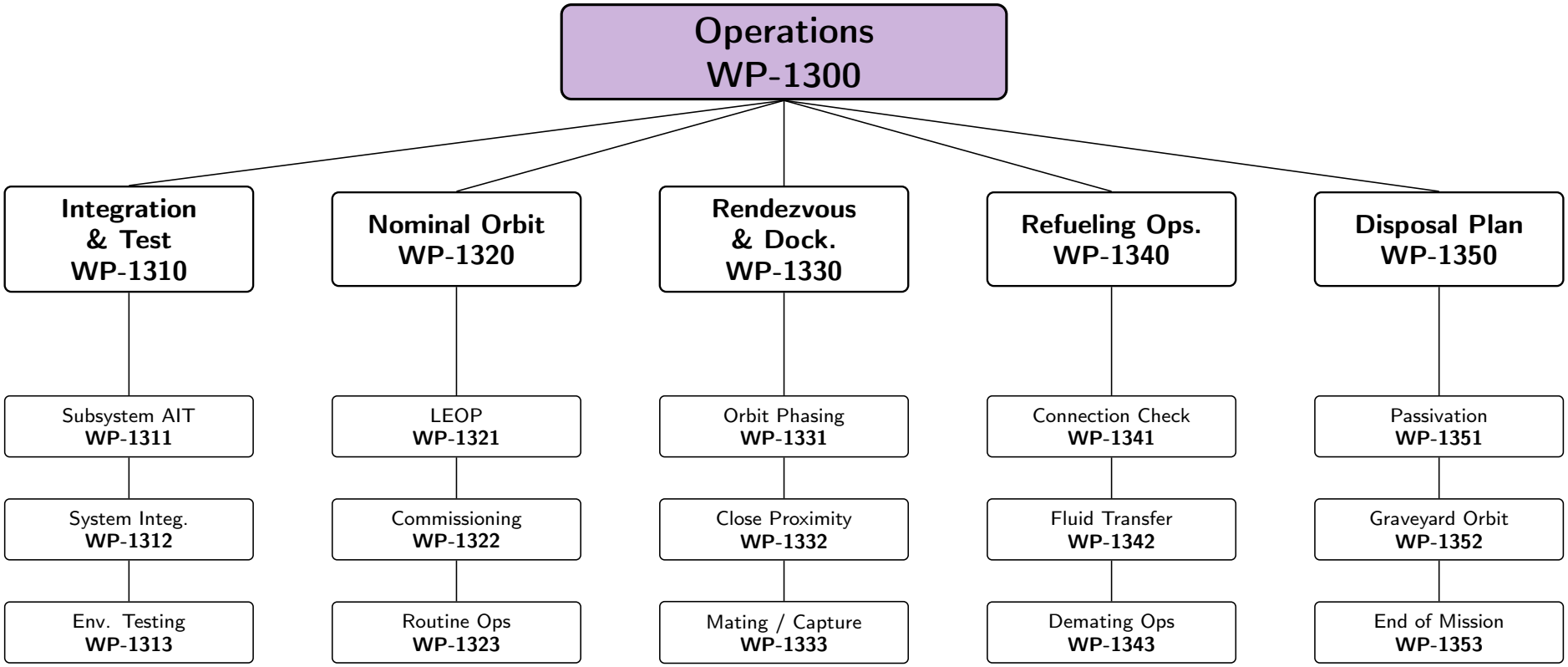


Figure 7.7: WBS of the Operations (WP-1300), down to Level 4.

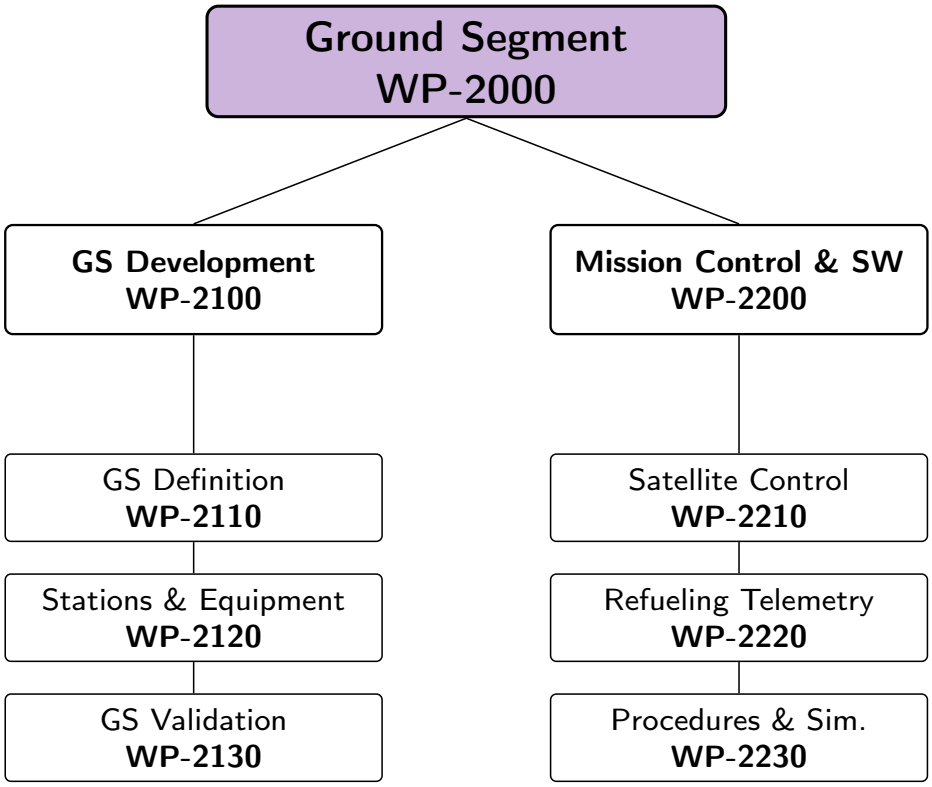


Figure 7.8: WBS of the Ground Segment (WP-2000).

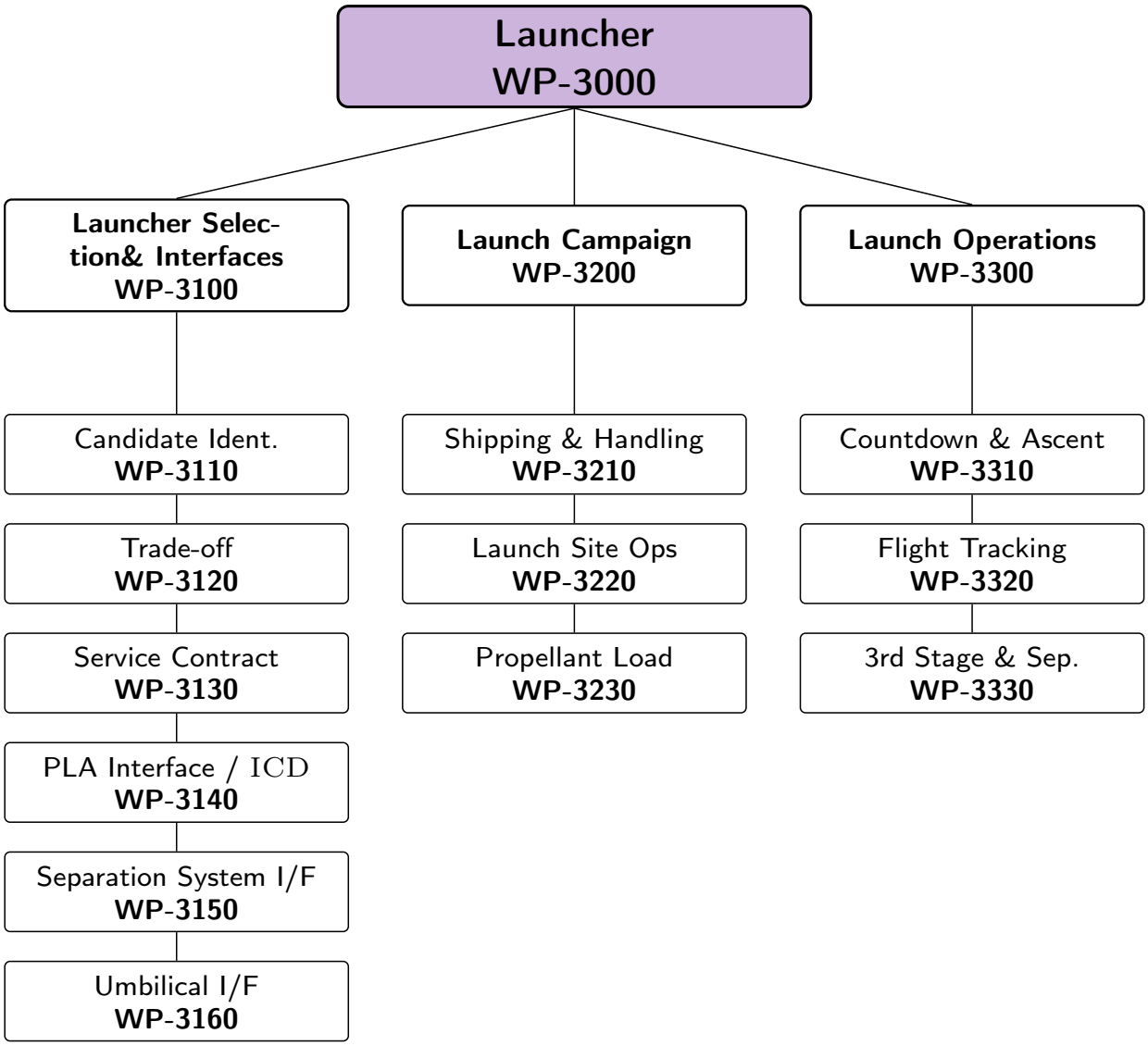


Figure 7.9: WBS of the Launcher (WP-3000).

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